Stochastic Analysis

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Preface

These notes accompany a sequel to my earlier study of stochastic processes, and grow out of the course on *Stochastic Analysis* taught by Prof. Jieliang HONG at SUSTech. After completing my exchange semester at SUSTech in 2025 Fall, I returned to my home institution. Yet each Wednesday I still travel to Shenzhen to sit in the lectures as an auditing student, and only later, often with some delay, rewrite the blackboard arguments into a form that can be revisited, tested, and refined.

If my first notebook began as a struggle to keep pace with martingales, Markov structure, and Brownian motion, this one begins at the point where the continuum becomes genuinely unforgiving: the moment we try to integrate with respect to a path of infinite variation. Stochastic analysis, to me, is the discipline of rebuilding real analysis under randomness, first by learning exactly *why* the Riemann-Stieltjes intuition fails for Brownian motion, and then by replacing it with Ito integration, quadratic variation, and the semimartingale viewpoint.

This is the mathematical backbone behind modern financial mathematics: the insistence that "fair" price dynamics should behave like martingales, and trading strategies should be modeled by predictable integrands whose stochastic integrals preserve that structure. The primary references for this notebook are the classic texts of Karatzas Shreve and Rogers Williams. I will follow their standards of rigor while keeping the narrative as close as possible to what I found most helpful when learning.

Finally, I am grateful to Prof. HONG for a lecture style that makes difficult concepts feel approachable without sacrificing precision. Any errors, omissions, or infelicities in exposition are entirely my own. To the reader who is also returning to these topics after feeling overwhelmed the first time: may these notes offer a steadier path into Ito calculus, and a reminder that the beauty of stochastic analysis lies not merely in its abstraction, but in its ability to turn uncertainty into structure.

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Chapter 0

Motivation

To initiate, recall the **Riemann-Stieltjes** Integral:

Definition 0.0.1. If f is continuous and g is of **finite variation**, then:

$$\int_a^b f(x)\mathrm{d}g(x)$$

is well defined to be, with $a \leq x_0 < x_1 < \cdots < x_n \leq b$ and $\xi_i \in [x_i, x_{i+1}]$:

$$\lim_{n\to\infty} \sum_{i=0}^{n-1} f(\xi_i)\left(g(x_{i+1}) - g(x_i)\right)$$

, and known as **Riemann-Stieltjes** Integral.

Then the main goal of stochastic analysis is to re-walk through the real analysis by rigorously defining the **Ito Integral** even for **stochastic processes**:

$$(H \cdot X)_t = \int_0^t H_s \mathrm{d}X_s$$

The original rule for Riemann-Stieltjes fails if we choose integrator to be Brownian motion.

Theorem 0.0.1. *For a standard brownian motion,* $\mathsf{B} = \{\mathsf{B}_t, t \geq 0\}$*:*

$$\sum_{k=1}^{2^n} \left(\mathsf{B}_{\frac{k}{2^n}} - \mathsf{B}_{\frac{k-1}{2^n}}\right)^2 \xrightarrow{a.s.} 1$$

Proof. On dyadic partition, for each fixed n, define, $\forall\ 1 \leq k \leq 2^n$:

$$\Delta_k^{(n)} := \mathsf{B}_{k/2^n} - \mathsf{B}_{(k-1)/2^n} \implies Q_n := \sum_{k=1}^{2^n} \left(\mathsf{B}_{k/2^n} - \mathsf{B}_{(k-1)/2^n}\right)^2 = \sum_{k=1}^{2^n} \left(\Delta_k^{(n)}\right)^2$$

Since B is a SBM, the random variables $\Delta_1^{(n)}, \ldots, \Delta_{2^n}^{(n)}$ are independent and

$$\Delta_k^{(n)} \sim N(0, 2^{-n}) \implies \mathsf{E}\left(\Delta_k^{(n)}\right)^2 = 2^{-n}$$

Therefore, the expectation of Q_n:

$$\mathsf{E}Q_n = \sum_{k=1}^{2^n} \mathsf{E}\left(\Delta_k^{(n)}\right)^2 = 2^n \cdot 2^{-n} = 1$$

Besides, by simple manipulation, the variance is given by, using independence:

$$\mathrm{Var}(Q_n) = \mathsf{E}\left(Q_n - 1\right)^2 = \sum_{k=1}^{2^n} \mathrm{Var}\left[\left(\Delta_k^{(n)}\right)^2\right] = 2^n \cdot 2 \cdot 2^{-2n} = 2^{1-n} \longrightarrow 0$$

Now fix $\varepsilon > 0$, by Chebyshev's inequality,

$$\mathsf{P}\big(|Q_n - 1| > \varepsilon\big) \leq \frac{\mathsf{E}\left(Q_n - 1\right)^2}{\varepsilon^2} = \frac{2^{1-n}}{\varepsilon^2} \implies \sum_{n=1}^{\infty} \mathsf{P}\big(|Q_n - 1| > \varepsilon\big) \leq \frac{1}{\varepsilon^2} \sum_{n=1}^{\infty} 2^{1-n} < \infty$$

By the Borel–Cantelli lemma,

$$\mathsf{P}\big(|Q_n - 1| > \varepsilon \text{ i.o.}\big) = 0.$$

Since this holds for every $\varepsilon = 1/m$, $m \in \mathbb{N}$, we conclude that

$$Q_n \to 1 \qquad \text{a.s.}$$

This completes the proof. $\square$

Using above easy-to-prove theorem, we first to be precise about **finite variation**:

Definition 0.0.2. The **Finite Variation** of a funtion $g : [a, b] \mapsto \mathbb{R}$ is defined to be:

$$V_g([a, b]) = \int_a^b |\mathrm{d}g(x)| = \sup_n \sum_{i=1}^{n} |g(x_{i+1}) - g(x_i)|$$

Now, we could justify that brownian motion fails in having infinite variation:

Corollary 0.0.2.

$$V_{\mathsf{B}}([0, 1]) = \infty, \ \ a.s.$$

Proof. Firstly, we notice the following inequality:

$$\begin{aligned}\sum_{k=1}^{2^n} \left(\mathsf{B}_{\frac{k}{2^n}} - \mathsf{B}_{\frac{k-1}{2^n}}\right)^2 &\leq \sup_{1 \leq k \leq 2^n} \left|\mathsf{B}_{\frac{k}{2^n}} - \mathsf{B}_{\frac{k-1}{2^n}}\right| \cdot \sum_{k=1}^{2^n} \left|\mathsf{B}_{\frac{k}{2^n}} - \mathsf{B}_{\frac{k-1}{2^n}}\right| \\ &\leq V_{\mathsf{B}}([0, 1]) \cdot \sup_{1 \leq k \leq 2^n} \left|\mathsf{B}_{\frac{k}{2^n}} - \mathsf{B}_{\frac{k-1}{2^n}}\right|\end{aligned}$$

Then if $V_{\mathsf{B}}([0, 1]) < \infty$, then by continuity of B, we have:

$$\sum_{k=1}^{2^n} \left(\mathsf{B}_{\frac{k}{2^n}} - \mathsf{B}_{\frac{k-1}{2^n}}\right)^2 \longrightarrow 0$$

, which is a contradiction, so $V_{\mathsf{B}}([0, 1]) = \infty$ with probability 1. $\square$

However, Ito found that, if we put some restriction on integrand and integrator, such as:

$$\begin{aligned} H_s &= \text{previsible/ predictable} \\ X_s &= \text{semimartingale} \end{aligned} \implies (H \cdot X)_t = \int_0^t H_s \mathrm{d}X_s \text{ is well defined.}$$

Example 0.0.1. Let $H_n \in \mathcal{F}_{n-1}$, and $M = \{M_n, n \in \mathbb{N}\}$ be a martingale, then:

$$(H \cdot M)_n = \sum_{k=1}^{n} H_k (M_k - M_{k-1})$$

is a martingale, which is known as discrete **martingale transformation**.

Besides rigorously defining the integral for general case, we also want the result integration keeps some desirable property, like **martingale**, this intuition comes from the financial application, where if the integral can't keep martingale, we could replicate an **arbitrage portfolio**. Then next section is to introduce some toolkits and also know the definition of **predictable, semimartingale**.

Chapter 1

Preliminary

Given a probability space $(\Omega, \mathcal{F}, \mathsf{P})$, a stochastic process $X = \{X_t, t \geq 0\}$ is a collection of r.v.s taking values in $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ formally, it is a **measurable function**:

$$\begin{aligned} X :&(t, \omega) \mapsto X_t(\omega) \\ &[0, \infty) \times \Omega \mapsto \mathbb{R}^d \end{aligned}$$

1.1 Sample Paths

Now if we fix ω, then the function:

$$t \mapsto X_t(\omega)$$

is called a **sample path**. Based on this concept, we could provide some useful definitions:

Definition 1.1.1. For two stochastic processes X and Y, we say Y is a **modification** of X if $\forall\, t \geq 0$, $\mathsf{P}(X_t = Y_t) = 1$

Remark 1.1.1. To be more precise, $\mathsf{P}(\{\omega : X_t(\omega) = Y_t(\omega)\}) = 1$

Most of time, modification is enough for us to inherit some property like continuity, but in order for the smoothness of knowledge and to get better understanding, we present a "slightly" stronger concept:

Definition 1.1.2. We say X and Y are **indistinguishable** if

$$\mathsf{P}\left(X_t = Y_t, \forall\, t \geq 0\right) = 1$$

Then Y is a **version** of X.

The reason why we say it is a slightly stronger version, is that due to the countable addivity, we could guarantee all the rational points to be the same, and if we have the "continuity" of the sample path we could approximate other points by rational points, then achieve indistinguishable, as following example gives:

Example 1.1.1. Let Y be a modification of X and suppose X and Y both have a.s. **right continuous** sample paths, then X and Y are indistinguishable.

Definition 1.1.3. A function f is **right-continuous** if:

$$f(x) = f(x^+) = \lim_{t \downarrow x} f(t)$$

Proof. Since $\forall\, t \geq 0$, $\mathsf{P}(X_t \neq Y_t) = 0$, then we get:

$$\mathsf{P}\left(\bigcup_{q \in \mathbb{Q}^+} X_q \neq Y_q\right) \leq \sum_{q \in \mathbb{Q}^+} \mathsf{P}(X_q \neq Y_q) = 0 \implies \mathsf{P}\left(\bigcap_{q \in \mathbb{Q}^+} X_q = Y_q\right) = 1$$

For two **right-continuous** functions f, g if:

$$f(q) = g(q), \forall\, q \in \mathbb{Q}^+ \implies f(t) = g(t), \forall\, t \geq 0$$

Therefore, using this result from **measure theory**, we land in:

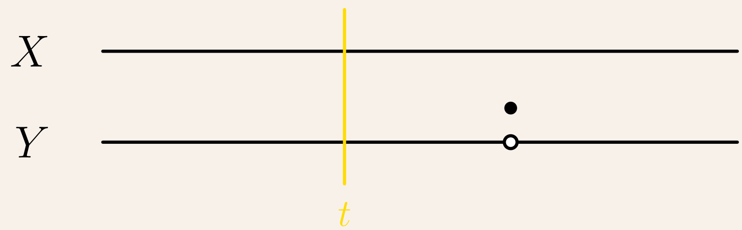
$$\mathsf{P}\left(X_t = Y_t, \forall\, t \geq 0\right) = \mathsf{P}\left(\bigcap_{t \geq 0} X_t = Y_t\right) = 1$$

Hence, we completed the proof. □

Please notice from above example 1.1.1 that we restrict to a.s. right-continuous sample path, so that we could land in approximation, but the following opposite example shows us the difference of modification and indistinguishable process.

Example 1.1.2. Let $T \sim \text{Exp}(1)$, define $X_t \equiv 0, \forall\, t \geq 0$, and another one:

$$Y_t = \begin{cases} 0, & t \neq T \\ 1, & t = T \end{cases}$$



Then we know:

$$\forall\, t \geq 0, \mathsf{P}(X_t \neq Y_t) = \mathsf{P}(Y_t = 1) = \mathsf{P}(t = T) = 0$$

so X is a modification of Y but **not a version of**, which you should verify.

1.2 Filtration

After talking about process, as processes go the information also increases, then to model this intuitive behavior, we introduce **filtration**: Let $\{\mathcal{F}_t, t \geq 0\}$ be a filtration of $(\Omega, \mathcal{F})$ such that $\forall\ 0 \leq s < t < \infty$:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$$

Typically, we just adapt to the so called **natural filtration**:

Example 1.2.1. Let $X = \{X_t, t \geq 0\}$ be a stochastic process, define:

$$\mathcal{F}_t^X = \sigma\left(X_s, 0 \leq s \leq t\right)$$

, then $\{\mathcal{F}_t^X, t \geq 0\}$ is a filtration.

From definition we could see, natural filtration is the minimum filtration that contains the information of the corresponding process, and also the adaptivity, i.e. $X_t \in \mathcal{F}_t^X$.

Example 1.2.2. Let X be a process, every sample paths of which is **RCLL**, let:

$$A = \{\omega : X(\omega) \text{ is continuous on } [0, t_0)\}$$

for some fixed $t_0 > 0$, then $A \in \mathcal{F}_{t_0}^X$.

Definition 1.2.1. The sample path of a process is **RCLL** if its sample path is **Right-Continuous with Left Limit**, i.e.:

$$f(x) = f(x^+),\ \ f(x^-)\ \exists$$

Proof. Since we already have right continuous, then what remain is show it meets with left limit:

$$A = \{\omega : \forall\ s \in (0, t_0), X_{s^-}(\omega) = X_s(\omega)\}$$

Similar trick, to rewrite event A into "rational number version":

$$A = \{\omega : \forall\ q \in (0, t_0) \cap \mathbb{Q}, X_{q^-}(\omega) = X_q(\omega)\}$$

It boils down into countable cases, to numerate $(0, t_0) \cap \mathbb{Q} = \{q_m, m \geq 1\}$:

$$\begin{aligned} A &= \left\{\omega : \forall\ q_m, \forall\ k \geq 1, \exists\ N \geq 1, \forall\ n \geq N, \left|X_{q_m - \frac{1}{n}}(\omega) - X_{q_m}(\omega)\right| < \frac{1}{k}\right\} \\ &= \bigcap_{m=1}^{\infty} \bigcap_{k=1}^{\infty} \bigcup_{N=1}^{\infty} \bigcap_{n=N}^{\infty} \left\{\left|X_{q_m - \frac{1}{n}}(\omega) - X_{q_m}(\omega)\right| < \frac{1}{k}\right\} \end{aligned}$$

Notice each $\left\{\left|X_{q_m - \frac{1}{n}}(\omega) - X_{q_m}(\omega)\right| < \frac{1}{k}\right\} \in \mathcal{F}_{t_0}^X$, then $A \in \mathcal{F}_{t_0}^X$. $\square$

Notice that example 1.2.2 requires every sample path to be RCLL, but if we lessen a little bit, even into a.s. RCLL, the conclusion may fail as following example suggests:

Example 1.2.3. Let X be a process whose sample paths are **RCLL** a.s., with all the same setup as example 1.2.2, then it is possible that $A \notin \mathcal{F}^X_{t_0}$.

Proof. WLOG, let $t_0 = 1$, and U be a non-measurable r,v. such that:

$$\mathsf{P}(U = 1) = 0$$

Define our X by:

$$X_t(\omega) = \begin{cases} 0, & U(\omega) \neq 1 \\ \begin{cases} 0, & t \neq \frac{1}{2} \\ 1, & t = \frac{1}{2} \end{cases}, & U(\omega) = 1 \end{cases}$$

By our construction, we know that:

$$A = \{\omega : X(\omega) \text{ is continuous on } [0, t_0)\} = \Omega / \{U = 1\}$$

, so if $\{U = 1\} \notin \mathcal{F}^X_{t_0}$, then $A \notin \mathcal{F}^X_{t_0}$. $\square$

From example 1.2.3, we see the key of the proof is to construct a zero-measure set such that it is not in our natural filtration, which is too tricky and hopefully be forbidden, then it gives us the motivation to consider broader class of filtration:

Definition 1.2.2. A filtration $\{\mathcal{F}_t, t \geq 0\}$ is said to satisfy the **usual condition** if:

(a) **Right-continuous**: $\mathcal{F}_t = \mathcal{F}_{t+} := \bigcap_{s>t} \mathcal{F}_s$.

(b) $\mathcal{F}_0$ contains all the **P-negligible** set in $\mathcal{F}$.

To complete the gap, we further provide the definitions of two:

Definition 1.2.3. An event A is **P-negligible** if $\exists\, N \in \mathcal{F}$, s.t. $A \subseteq N, \mathsf{P}(N) = 0$.

Another one is "left/right limit" of filtration:

Definition 1.2.4. We say a filtration is **right-continuous** if $\mathcal{F}_t = \mathcal{F}_{t+}$, where:

$$\mathcal{F}_{t+} = \bigcap_{s>t} \mathcal{F}_s, \mathcal{F}_{t-} = \sigma\left(\bigcup_{s<t} \mathcal{F}_s\right)$$

After introducing information flow, we need to relate it with process:

Definition 1.2.5. The stochastic process X is **adapted** to the filtration $\{\mathcal{F}_t, t \geq 0\}$ if $\forall\, t \geq 0$, $X_t \in \mathcal{F}_t$. Clearly, X is adapted to the **natural filtration**.

Adaptivity is too loose, and impedes understanding, thus we consider **progressively measurable**.

Definition 1.2.6. The stochastic process X is called **progressively measurable** w.r.t $\{\mathcal{F}_t, t \geq 0\}$ if $\forall\, t \geq 0$, the map following is measurable.

$$(s,\omega) \mapsto X_s(\omega) : ([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \mapsto \left(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)\right)$$

Remark 1.2.1. To compare, adaptivity is to require measurability to:

$$(s,\omega) \mapsto X_s(\omega) : ([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}) \mapsto \left(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)\right)$$

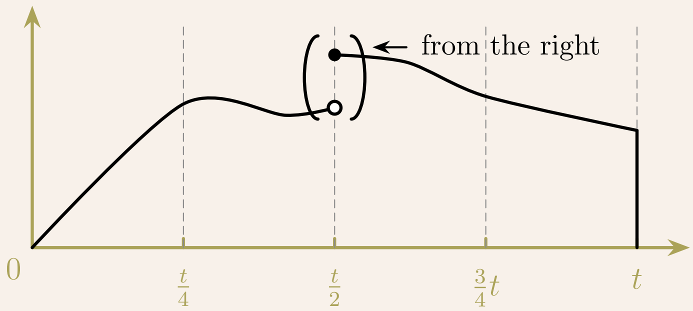
Intuitively speaking, progressively measurable is to require the information is "strictly" enough at this each time, to see the relations:

$$\text{adapted} \supseteq \text{progressively measurable} \supseteq \text{previsible}$$

Most of cases, we don't differentiate them, since it is really close as below example shows:

Example 1.2.4. If X is adapted to filtration $\{\mathcal{F}_t, t \geq 0\}$ and every sample path is right-continuous (same for left-continuous), then X is also progressively measurable.

Proof. Since right continuity, $\forall\, t > 0, \forall\, n \geq 1, \forall\, 0 \leq k \leq 2^n - 1$, we construct:



$$X_s^{(n)}(\omega) = X_{\frac{k+1}{2^n}t}(\omega), \forall\, \frac{kt}{2^n} < s \leq \frac{(k+1)t}{2^n}$$

, then $\lim_{n\to\infty} X_s^{(n)}(\omega) = X_s(\omega)$ for all $(s,\omega) \in [0,t] \times \Omega$. Recall that limit perserve our measurability, then it suffices to prove:

$$(s,\omega) \mapsto X_s^{(n)}(\omega) : ([0,t] \times \Omega, \mathcal{B}([0,t]) \otimes \mathcal{F}_t) \mapsto \left(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)\right)$$

is measurable, however, this is clearly true since, $\forall\, A \in \mathcal{B}(\mathbb{R}^d)$:

$$\{(s,\omega) : X_s^{(n)}(\omega) \in A\} = \bigcup_{k=0}^{2^n-1} \left(\frac{k}{2^n}t, \frac{k+1}{2^n}t\right] \times \left\{X_{\frac{k+1}{2^n}t} \in A\right\} \in \mathcal{B}([0,t]) \otimes \mathcal{F}_t$$

Therefore, we completed the proof. $\square$

1.3 Stopping Times

Stopping times are really important subject of interest, since we already known that original brownian motion has infinite variation, but if we use stopping times to replace the original time, then to construct **local martingale**, which can be used as integrator.

Definition 1.3.1. A random time T is a **stopping time** of $\{\mathcal{F}_t, t \geq 0\}$ if $\forall\, t \geq 0$, $\{T \leq t\} \in \mathcal{F}_t$, another one is **optional time** if $\forall\, t \geq 0$, $\{T < t\} \in \mathcal{F}_t$.

Intuitively speaking, stopping time is a r.v. such that its property can be checked at each time (i.e. the information ($\{T \leq t\}$) of it contains within time t ($\mathcal{F}_t$)), and optional time is by time t but not included. However, they don't have too much difference as below proposition suggests:

Proposition 1.3.1. *If T is a stopping time, then also a optional time. Conversely, if the filtration is **right-continuous**, then stopping time is same as optional time.*

Proof. First one is easy to see, we focus on latter equivalence statement:

$\Longrightarrow$ The forward direction is simple, as we notice below:

$$\{T < t\} = \bigcup_{n=1}^{\infty} \left\{T \leq t - \frac{1}{n}\right\}$$

, as each $\{T \leq t - 1/n\} \in \mathcal{F}_{t-1/n} \subseteq \mathcal{F}_t$, then $\{T < t\} \in \mathcal{F}_t$.

$\Longleftarrow$ For backward direction, it is non-trival, $\forall\, m \geq 1$:

$$\{T \leq t\} = \bigcap_{n=m}^{\infty} \left\{T < t + \frac{1}{n}\right\}$$

, as each $\{T < t + 1/n\} \in \mathcal{F}_{t+1/n} \subseteq \mathcal{F}_{t+1/m}$, then $\{T \leq t\} \in \mathcal{F}_{t+1/m}$.

Lastly, we finished the proof by the arbitrariness of m, then $\{T \leq t\} \in \mathcal{F}_{t+} = \mathcal{F}_t$. $\square$

Next, after we defined the stopping time, let us see some properties:

Proposition 1.3.2. *If T and S are stopping times, then:*

$$T \wedge S,\ T \vee S,\ T + S$$

are all stopping times.

Proof. The first two are easy once we seperate the event properly:

(1) For $T \wedge S$, we could consider as below:

$$\{T \wedge S \leq t\} = \{T \leq t\} \bigcup \{S \leq t\} \in \mathcal{F}_t$$

, as T, S are stopping times, then $\{T \leq t\}, \{S \leq t\} \in \mathcal{F}_t$.

(2) For $T \vee S$, we do the similar seperation, $\{T \vee S \leq t\} = \{T \leq t\} \bigcap \{S \leq t\}$.

(3) $T+S$ is tricky, where we should transfer using closure under complement:

$$\{T+S \leq t\} \in \mathcal{F}_t \Longleftrightarrow \{T+S > t\} \in \mathcal{F}_t$$

After we do this transformation, we could seperate this by:

$$\{T+S > t\} = \{T=0, S>t\} \bigcup \{S=0, T>t\} \bigcup \{T \geq t, S>0\} \bigcup \{0<T<t, T+S>t\}$$

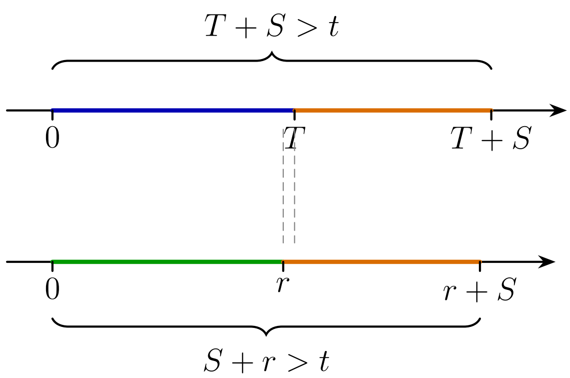
For the first one, it is easy to see:

$$\{T=0, S>t\} = \{T=0\} \bigcap \{S>t\} = \{T=0\} \bigcap \{S \leq t\}^c \in \mathcal{F}_t$$

Then using similar method, we could easily check middle two, what remains is:

Lemma.

$$\{0<T<t, T+S>t\} = \bigcup_{\substack{0<r<t \\ r \in \mathbb{Q}^+}} \{r<T<t, S>t-r\}$$



Proof. **of Lemma:**

$\Longleftarrow$ Backward direction is easy, as $S > t-r > t-T$, hence $S+T>t$.

$\Longrightarrow$ Forward direction relies on the dense of rational number, where since $S+T>t$, then by dense of rational numbers, $\exists\, r \in (0,T)$, s.t. $S+r>t$.

Last one can be visualised as above by moving r really close to T. □

With the lemma above, we could seperate event as before, then complete the proof. □

Lastly, if we have a set of stopping times, then the sup of it is also a stopping time:

Proposition 1.3.3. *If $\{T_n, n \geq 1\}$ are s.t., then $\sup_{n\geq 1} T_n$ is a s.t.*

Proof. This proof is simple and straightforward, $\forall\, t \geq 0$:

$$\left\{\sup_{n\geq 1} T_n \leq t\right\} = \bigcap_{n=1}^{\infty} \{T_n \leq t\} \in \mathcal{F}_t$$

Then we complete the proof. □

Since we already have s.t. as a kind of time, it is reasonable to ask its σ algebra:

Definition 1.3.2. The stopping σ-algebra is:

$$\mathcal{F}_T = \left\{ A \in \mathcal{F} : \forall\, t \geq 0, A \bigcap \{T \leq t\} \in \mathcal{F}_t \right\}$$

Intuitively speaking, above definition 1.3.2 is to say for every event in stopping σ algebra, given the time t we are at, we could know whether it happens or not.

Proposition 1.3.4. *If $T \equiv t_0$, then $\mathcal{F}_T = \mathcal{F}_{t_0}$.*

Proof. Firstly, we observe that $\forall\, t \geq 0$:

$$\{T \leq t\} = \begin{cases} \emptyset, & t_0 > t \\ \Omega, & t_0 \leq t \end{cases}$$

$\Longleftarrow$ Then we firstly check the backward direction, $\forall\, A \in \mathcal{F}_{t_0}$, $\forall\, t \geq 0$:

(1) If $t \geq t_0$, then:

$$A \cap \{T \leq t\} = A \cap \Omega = A \in \mathcal{F}_{t_0} \subseteq \mathcal{F}_t$$

(2) If $t < t_0$, then:

$$A \cap \{T \leq t\} = A \cap \emptyset = \emptyset \in \mathcal{F}_t$$

Therefore, $A \in \mathcal{F}_T$ as definition states.

$\Longrightarrow$ Forward direction is similar, as $\forall\, A \in \mathcal{F}_T$, and $\forall\, t \geq 0$, then:

$$A \cap \{T \leq t\} \in \mathcal{F}_t \implies A = A \cap \Omega = A \cap \{t_0 \leq t_0\} = A \cap \{T \leq t_0\} \in \mathcal{F}_{t_0}$$

Therefore, we completed the proof. □

Besides, we will also be interested into the relationship of a set of s.t.s,

Proposition 1.3.5. *$\forall$ stopping times T and S, $\forall\, A \in \mathcal{F}_S$, we have:*

$$A \cap \{S \leq T\} \in \mathcal{F}_T$$

In particular, if $S \leq T$ for all $\omega \in \Omega$, then:

$$\mathcal{F}_S \subseteq \mathcal{F}_T$$

Proof. The key idea is to use the condition that $A \in \mathcal{F}_S$, so that:

$$A \cap \{S \leq t\} \in \mathcal{F}_t$$

And in order to show, $A \cap \{S \leq T\} \in \mathcal{F}_T$, we need $\forall\, t > 0$,

$$A \cap \{S \leq T\} \cap \{T \leq t\} \in \mathcal{F}_t$$

Notice here, since on $\{S \leq T\} \cap \{T \leq t\}$:

(a) $\{S \leq T\} \cap \{T \leq t\} = \{S \leq T \leq t\} \implies S \leq t$.

(b) $\{S \leq T\} = \{S \wedge t \leq T \wedge t\}$.

Then, it suffice to show, $\forall\, t > 0$:

$$(A \cap \{S \leq t\}) \cap \{T \leq t\} \cap \{S \wedge t \leq T \wedge t\} \in \mathcal{F}_t$$

The first lives in $\mathcal{F}_t$ followed from the fact that $A \in \mathcal{F}_S$, and second comes from the fact that T is a s.t., it remains to show, $\{S \wedge t \leq T \wedge t\} \in \mathcal{F}_t$:

Lemma. *Let T and S be two s.t.s, then $\forall\, t \geq 0$:*

$$\{S \wedge t \leq T \wedge t\} \in \mathcal{F}_t$$

Proof. of Lemma
This follows the similar structure of proposition 1.3.2's lemma to decompose:

$$\{S \wedge t \leq T \wedge t\} = \bigcup_{r \in \mathbb{Q}^+ \cap [0,t]} \{S \wedge t \leq r\} \cap \{T \wedge t \geq r\}$$

Using the closure of complement of σ-algebra to show its conclusion. □

Therefore, we completed the proof for general situation, and in particular, since $S \leq T$ for all $\omega \in \Omega$, then $\{S \leq T\} = \Omega$, so $\forall\, A \cap \Omega = A \in \mathcal{F}_S, A \in \mathcal{F}_T \implies \mathcal{F}_S \subseteq \mathcal{F}_T$. □

To reiterate, the reason why we introduce the stopped time $(T \wedge t)$ is to relate the original s.t. with t, so that we could manipulate on it with relationship pf $\mathcal{F}_t$.

Proposition 1.3.6. *Let S and T be two s.t.s, then:*

$$\mathcal{F}_{S \wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$$

, and also:

$$\{T < S\}, \{S < T\}, \{T \leq S\}, \{S \leq T\}, \{S = T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

Proof. The intuition is clear, for σ-algebra for s.t.s which is the smallest of two, is certainly the intersection of two σ-algebra of two s.t.s Let's prove them in order:

(a) Firstly, consider the inclusion:

$\Longrightarrow$ From proposition 1.3.5, we have:

$$\begin{cases} S \wedge T \leq S \\ S \wedge T \leq T \end{cases} \implies \begin{cases} \mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_S \\ \mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_T \end{cases} \implies \mathcal{F}_{S \wedge T} \subseteq \mathcal{F}_S \cap \mathcal{F}_T$$

$\Longleftarrow$ For backward direction, $\forall\, A \in \mathcal{F}_S \cap \mathcal{F}_T$, $\forall\, t > 0$, we need to show:

$$A \cap \{S \wedge T \leq t\} = A \cap (\{S \leq t\} \cup \{T \leq t\}) \in \mathcal{F}_t$$

, above is easy to show by distribution rule and the fact that $A \in \mathcal{F}_S, \mathcal{F}_T$:

$$A \cap (\{S \leq t\} \cup \{T \leq t\}) = (A \cap \{S \leq t\}) \cup (A \cap \{T \leq t\}) \in \mathcal{F}_t$$

(b) Similarly, from proposition 1.3.5, we have, $\forall\ A \in \mathcal{F}_S$, we have:

$$A \cap \{S \leq T\} \in \mathcal{F}_T \overset{\text{Choose } A=\Omega}{\Longrightarrow} \{S \leq T\} \in \mathcal{F}_T \implies \{S \leq T\}^c = \{S > T\} \in \mathcal{F}_T$$

It remains to show, $\{S < T\} \in \mathcal{F}_T$, since others can be shown by interchanging the place of them, which we will see it later, here notice that, in order to introduce the relations of S and T, we construct a stopped s.t. by defining $R := S \wedge T$, then:

$$\{S < T\} = \{R < T\}, R \in \mathcal{F}_R \subseteq \mathcal{F}_T \implies \{S < T\} = \{R < T\} \in \mathcal{F}_T$$

Lastly, we use the power of interchanging the orders,

$$\{T \leq S\}, \{T > S\}, \{S > T\} \in \mathcal{F}_S$$

, so that other parts can all be expressed by their complements:

(i) For $\{S \leq T\} \in \mathcal{F}_T$, we use $\{S > T\} \in \mathcal{F}_S$:

$$\{S \leq T\} = \{S > T\}^c \in \mathcal{F}_S \implies \{S \leq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

(ii) For $\{S < T\} \in \mathcal{F}_T$, we use $\{T \leq S\} \in \mathcal{F}_S$:

$$\{S < T\} = \{T \geq S\}^c \in \mathcal{F}_S \implies \{S < T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

(iii) Lastly, for $\{T = S\}$, we use (i) and (ii) $\{S \geq T\} = \{S < T\}^c \in \mathcal{F}_S \cap \mathcal{F}_T$:

$$\{S = T\} = \{S \leq T\} \cap \{S \geq T\} \in \mathcal{F}_S \cap \mathcal{F}_T$$

Then we completed the proof. □

Remark 1.3.1. A short summary of the properties for s.t.s:

(1) For s.t. T, $T \in \mathcal{F}_T$, i.e. $\forall\ t > 0$, $\{T < t\} \in \mathcal{F}_T$.

(2) For two s.t.s, S and T, if $S \leq T$, then $\mathcal{F}_S \subseteq \mathcal{F}_T$.

(3) For s.t. T, and $\forall\ t > 0$, then $T \wedge t \in \mathcal{F}_{T\wedge t} \subseteq \mathcal{F}_t$.

(4) For two s.t.s S and T, then $\mathcal{F}_{S\wedge T} = \mathcal{F}_S \cap \mathcal{F}_T$.

Lastly, we need to justify its validity of changing time with s.t.:

Theorem 1.3.7. *Let $X = \{X_t, t \geq 0\}$ be a progressively measurable process, and T be a s.t., then X_T defined on $\{T < +\infty\}$ is $\mathcal{F}_T$-measurable. And the stopped process $\{X_{T\wedge t}, t \geq 0\}$ is also progressively measurable.*

Proof. The two statements is equivalent, since if we want to show, X_T defined on $\{T < +\infty\}$ is $\mathcal{F}_T$-measurable, then it is equal to show, $\forall\ t \geq 0$, and $\forall\ B \in \mathcal{B}(\mathbb{R}^d)$:

$$\{X_T \in B\} \cap \{T \leq t\} \in \mathcal{F}_t \iff \{X_{T\wedge t} \in B\} \cap \{T \leq t\} \in \mathcal{F}_t$$

Then it remains to show, $X_{T\wedge t} \in \mathcal{F}_t$, by observing:

$$(s, \omega) \mapsto (T(\omega) \wedge s, \omega)$$

is $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable, and since the process is progressively measurable:

$$(s,\omega) \mapsto X_s(\omega)$$

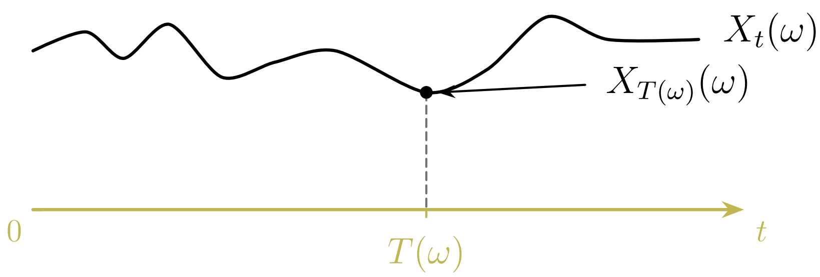
is $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable, then by composition rule:

$$(s,\omega) \mapsto (T(\omega) \wedge s, \omega) \mapsto (X_{T(\omega)\wedge s}, \omega)$$

is also $\mathcal{B}([0,t]) \otimes \mathcal{F}_t$-measurable, then we completed the proof. $\square$

Remark 1.3.2. To remind, we interpret the stochastic process replaced time with s.t. (or random time), by following composite functions and figure:

$$\omega \mapsto (T(\omega), \omega) \mapsto X_{T(\omega)}(\omega)$$



Chapter 2

Continuous-time Martingale

Martingale sits at an important role of this course, and also stochastic process, for this session, we mainly focus on how to transfer the classical discrete time martingale properties (where most of reader should learn from classical stochastic process like Rick. Durrett's book) to continuous time version, and most powerful tool for proving convergence.

Definition 2.0.1. A martingale is an adapted stochastic process $X = \{X_t, t \geq 0\}$ such that:

(a) $\forall\, t \geq 0, \mathsf{E}|X_t| < \infty$.

(b) $\forall\, t \geq s \geq 0, \mathsf{E}\left(X_t \mid \mathcal{F}_s\right) = X_s$

Remark 2.0.1. A simple extension to sub/super-martingale:

(1) If definition 2.0.1's (b) change to:

$$\mathsf{E}\left(X_t \mid \mathcal{F}_s\right) \geq X_s$$

, then it is called submartingale.

(2) If definition 2.0.1's (b) change to:

$$\mathsf{E}\left(X_t \mid \mathcal{F}_s\right) \leq X_s$$

, then it is called supermartingale.

2.1 Martingale Theory in Nutshell

Following we presents some important theorem some of them without proof:

Theorem 2.1.1. *Let $\varphi : \mathbb{R} \mapsto \mathbb{R}$ be a convex (concave up) function, if $X = \{X_t, t \geq 0\}$ is a martingale, then $\varphi(X) = \{\varphi(X_t), t \geq 0\}$ is a submartingale.*

Proof. By Jensen's inequality:

$$|\mathsf{E}X_t| \leq \mathsf{E}\,|X_t| \implies |\mathsf{E}\left(X_t \mid \mathcal{F}_s\right)| \leq \mathsf{E}\left(|X_t| \mid \mathcal{F}_s\right)$$

Hence, by assuming integrability and adaptivity, it remains:

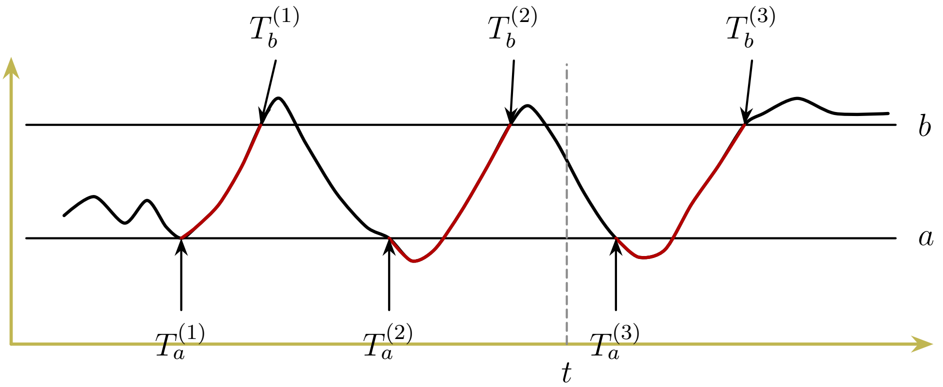
$$\mathsf{E}\left(\varphi(X_t) \mid \mathcal{F}_s\right) \geq \varphi\left(\mathsf{E}\left(X_t \mid \mathcal{F}_s\right)\right) = \varphi(X_s)$$

Then we completed the proof. □

Remark 2.1.1. Some common convex function we used are, where $p \geq 1$.

$$|X_t|, \quad X_t^2, \quad (X_t)^+ = \max(X_t, 0), \quad |X_t|^p$$

Next important theorem is **continuous-time Doob's upcrossing inequality**:



As the figure shows, we could denote the i th upcrossings (or downcrossings) time to be the first time, after the i-1 th downcrossings (or upcrossings), more formally:

$$\begin{aligned}
T_b^0 &= -\infty \\
T_a^{(1)} &= \inf\left\{t \geq \left(T_b^{(0)}\right)^+ : X_t \leq a\right\} \\
T_b^{(1)} &= \inf\left\{t \geq T_a^{(1)} : X_t \geq b\right\}
\end{aligned}$$

After the initial setup, for $\forall\, \ell \geq 2$:

$$\begin{aligned}
T_a^{(\ell)} &= \inf\left\{t \geq T_b^{(\ell-1)} : X_t \leq a\right\} \\
T_b^{(\ell)} &= \inf\left\{t \geq T_a^{(\ell-1)} : X_t \geq b\right\}
\end{aligned}$$

Then we focus on the # of upcrossing in an interval $[0, t]$ for given two bounds ($a < b$):

$$U_{ab}^X[0, t] = \max\left\{k \geq 0 : T_b^{(k)} \leq n\right\}$$

Now, we could present our most important theorem about submartingale:

Theorem 2.1.2. ***(Continuous-time Doob's Upcrossings Inequality)*** *Let $X = \{X_t, t \geq 0\}$ be a right-continuous submartingale, then for all $T > 0$:*

$$\mathsf{E}U_{ab}^X\left[0, T\right] \leq \frac{\mathsf{E}\left(X_T - a\right)^+}{b - a}$$

Proof. The idea is to how to apply the discrete-time Doob's upcrossing inequality, where we assume to be known, since we presume all readers have studied basic stochastic process. Then we could try to discretise the continuous time, for each $n \geq 1$, we let:

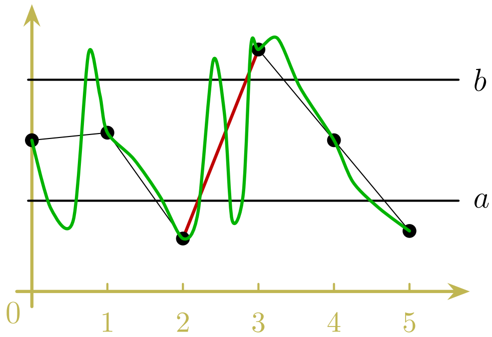
$$D_n = \mathbb{Z}/2^n = \left\{0, \frac{\pm 1}{2^n}, \frac{\pm 2}{2^n}, \frac{\pm 3}{2^n}, \cdots\right\}$$

Consider the upcrossings of X on $[0, T] \cap D_n$, where:

$$\{X_{j/2^n}, j \geq 0\}$$

, is **discrete-time submartingale**, then as showed as the figure below, we have:

$$U_{ab}^X\left([0, T] \cap D_n\right) \leq U_{ab}^X[0, T]$$



Clearly if we discretise it finer, it will be nondecreasing, then:

$$U_{ab}^X\left([0, T] \cap D_n\right) \uparrow U_{ab}^X[0, T]$$

Therefore, by using discrete-time upcrossing inequality, i.e.

$$U_{ab}^X\left([0, T] \cap D_n\right) \leq \frac{1}{b-a}\mathsf{E}\left(X_{T_n} - a\right)^+ \leq \frac{1}{b-a}\mathsf{E}\left(X_T - a\right)^+$$

, where $T_n := \frac{\lfloor T \cdot 2^n \rfloor}{2^n}$, let $n \to \infty$ by limiting theory, we finished the proof. □

Just as we said, it is powerful tool to show the convergence for submartingale:

Proposition 2.1.3. *If $X = \{X_t, t \geq 0\}$ is a right-continuous submartingale, and:*

$$\sup_{t \geq 0} \mathsf{E}X_t^+ < \infty$$

, then $\exists\ X \in L^1$ such that, as $t \longrightarrow \infty$:

$$X_t \xrightarrow{a.s.} X$$

Proof. Here, we ignore the procedure of showing existence, just focus on convergence:

$$\begin{aligned}
\mathsf{E}U_{ab}^X[0, \infty) &= \lim_{n \to \infty} \mathsf{E}U_{ab}^X[0, n] \\
&\leq \limsup_{n \to \infty} \frac{1}{b-a}\mathsf{E}\left(X_n - a\right)^+ \\
&\leq \limsup_{n \to \infty} \frac{1}{b-a}\left(\mathsf{E}X_n^+ + |a|\right) < \infty
\end{aligned}$$

, above is provided by theorem 2.1.2 and condition, hence:

$$\forall a < b, U_{ab}^X[0, , \infty) < \infty, \text{ a.s.} \quad \Longrightarrow \quad \text{w.p.1 } \forall\, a < b \in \mathbb{Q}, U_{ab}^X[0, \infty) < \infty$$

, this is given by the fact that countable additivity of our measure, therefore:

$$\liminf_{t\to\infty} X_t < a \leq b < \limsup_{t\to\infty} X_t$$

does not occur, which is equivalent to say:

$$\liminf_{t\to\infty} X_t = \limsup_{t\to\infty} X_t$$

, then we completed the proof. □

To connect relations of different convergence modes, we need one more definition:

Definition 2.1.1. A family of r.v.s $\{X_t, t \geq 0\}$ is **uniformly integrable** (u.i.) if:

$$\lim_{M\to\infty} \sup_{t\geq 0} \mathsf{E}|X_t|\mathbf{1}_{\{|X_t|>M\}} = 0$$

Remark 2.1.2. Using Markov Inequality, if $\sup_{t\geq 0} \mathsf{E}|X_t|^p < \infty$, for some $p > 1$, then X is u.i.

With **uniformly integrable**, we could present the following equvalence:

Theorem 2.1.4. *If $X_n \xrightarrow{p.} X$, then the following are equivalent:*

(a) $X_n \xrightarrow{L^1} X$.

(b) $\mathsf{E}|X_n| \longrightarrow \mathsf{E}|X|$.

(c) $\{X_n, n \geq 0\}$ *is u.i.*

Proof. We omit the proof since it is not our focus. □

Last important property is **last elements**, before that we need:

Proposition 2.1.5. *Let $Z \in L^1$, then following is u.i.*

$$\{\mathsf{E}\left(Z \mid \mathcal{G}\right) : \mathcal{G} \textit{ is a sub } \sigma\textit{-field of } \mathcal{F}\}$$

Proof. Proof omitted.

Remark 2.1.3. In particular, if we pick $\mathcal{G}$ to be filtration, i.e.:

$$\{\mathsf{E}\left(Z \mid \mathcal{F}_t\right), t \geq 0\}$$

, is u.i. and also a **martingale**.

□

With the help of u.i., we could show following relations, (last element), providing convenience in proving **Optional Sampling Theorem** (OST).

Proposition 2.1.6. *A right-continuous martingale $\{X_t, t \geq 0\}$ is u.i. iff:*

$$\exists\, X_\infty \in L^1, \text{ s.t. } X_t = \mathsf{E}\left(X_\infty \mid \mathcal{F}_t\right)$$

Proof. Firstly, we start with the inclusion:

$\Longrightarrow$ Firstly, by u.i., we have $\sup_{t\geq 0} \mathsf{E}|X_t| < \infty$, then through proposition 2.1.3:

$$\exists\, X_\infty, \text{ s.t. } X_t \xrightarrow{a.s.} X_\infty$$

, more important, we know it is L^1 since:

$$\mathsf{E}|X_\infty| = \mathsf{E} \lim_{t\to\infty} |X_t| \leq \mathsf{E} \liminf_{t\to\infty} \mathsf{E}|X_t| < \infty$$

, where last inequality comes from **Fatou's Lemma**, next to prove:

$$\forall\, t \geq 0, X_t = \mathsf{E}\left(X_\infty \mid \mathcal{F}_t\right)$$

is equivalent, by the definition of conditional expectation, to show $\forall\, A \in \mathcal{F}_t$:

$$\mathsf{E}X_t \mathbf{1}_A = \mathsf{E}X_\infty \mathbf{1}_A$$

Common as usual, we approximate it by sending limit to infinity, for any $s \geq 0$:

$$\mathsf{E}\left(X_{t+s} \mid \mathcal{F}_t\right) = X_t \implies \mathsf{E}X_{t+s}\mathbf{1}_A = \mathsf{E}X_t\mathbf{1}_A$$

For limiting validity, we recall that $u.i. + a.s. \implies L^1$:

$$|\mathsf{E}X_{t+s}\mathbf{1}_A - \mathsf{E}X_\infty\mathbf{1}_A| \leq \mathsf{E}\left|X_{t+s} - X_\infty\right| \longrightarrow 0$$

Then by sending $s \to \infty$ we complete the fisrt part of the proof.

$\Longleftarrow$ This direction immediately follows from proposition 2.1.5.

Then we completed the proof by showing both directions. □

After all the preparation, we commence the OST:

Theorem 2.1.7. ***(Optional Sampling Theorem)***
Let $\{X_t, t \geq 0\}$ be a right-continuous martingale, let $S \leq T$ be two s.t.s suppose:

(a) $\{X_t, t \geq 0\}$ *is u.i.*

(b) $\exists$ *constant* $K > 0$, *s.t.* $S, T \leq K$

, either above two satisfied, then we have:

$$\mathsf{E}\left(X_T \mid \mathcal{F}_S\right) = X_S$$

Proof. In order to use proposition 2.1.6, we consider:

(a) Then by proposition 2.1.3, we choose $Z = X_\infty$

(b) We directly pick the "last element", i.e. $Z = X_K$

In either case, we have: $X_t = \mathsf{E}(Z \mid \mathcal{F}_t)$, then it suffices to show:

$$X_T = \mathsf{E}(Z \mid \mathcal{F}_T)$$

Since provided the above satified, by symmetry, we could interchanged it to S, so that:

$$X_S = \mathsf{E}(Z \mid \mathcal{F}_S)$$

Hence, by tower rule and proposition 1.3.5, then:

$$\mathsf{E}(X_T \mid \mathcal{F}_S) = \mathsf{E}[\mathsf{E}(Z \mid \mathcal{F}_T) \mid \mathcal{F}_S] = \mathsf{E}(Z \mid \mathcal{F}_S) = X_S$$

Now, we could firstly consider the discrete cases:

(1) If T can be decomposed into discrete cases, i.e. $T \in \{t_1, t_2, \cdots\}$, then $\forall\ A \in \mathcal{F}_T$:

$$\begin{aligned} \mathsf{E}X_T\mathbf{1}_A = \mathsf{E}Z\mathbf{1}_A &= \sum_{n=1}^{\infty} \mathsf{E}X_{t_n}\mathbf{1}_{\{T=t_n\}}\mathbf{1}_A \\ &\text{Because } \begin{cases} A \cap \{T = t_n\} \in \mathcal{F}_{t_n} \\ \mathsf{E}(Z \mid \mathcal{F}_{t_n}) = X_{t_n} \end{cases} \\ &= \sum_{n=1}^{\infty} \mathsf{E}Z\mathbf{1}_{A\cap\{T=t_n\}} = \mathsf{E}Z\mathbf{1}_A \end{aligned}$$

(2) For general T, we discretise from right (for the sake of right-continuous), then define:

$$T_k = \frac{\lfloor 2^k \cdot T \rfloor + 1}{2^k}$$

, clearly $T_k \downarrow T$, then $\forall\ A \in \mathcal{F}_T \subseteq \mathcal{F}_{T_k}$:

$$\mathsf{E}(Z \mid \mathcal{F}_{T_k}) = X_{T_k} \implies \mathsf{E}Z\mathbf{1}_A = \mathsf{E}X_{T_k}\mathbf{1}_A$$

Sending the $k \to \infty$ to see $\mathsf{E}Z\mathbf{1}_A = \mathsf{E}X_T\mathbf{1}_A$, which is provided by:

$$\begin{cases} X_{T_k} \xrightarrow{a.s.} X_T \text{ by right-continuous} \\ \{X_{T_k}\} \text{ is u.i. by proposition 2.1.5} \end{cases} \implies X_{T_k} \xrightarrow{L^1} X_T$$

Hence, we completed the proof, by the help of dicretisation. □

By closing subsections for martingale, we presents some useful inequality:

Theorem 2.1.8. ***(Doob's Maximal Inequality)***
Let $X = \{X_t, t \geq 0\}$ be a ***right-continuous submartingale****, then $\forall\ \lambda > 0$:*

$$\mathsf{P}\left(\sup_{0\leq s\leq t} X_s > \lambda\right) \leq \frac{1}{\lambda}\mathsf{E}X_t^+$$

$$\mathsf{P}\left(\inf_{0\leq s\leq t} X_s < -\lambda\right) \leq \frac{1}{\lambda}\left(\mathsf{E}X_t^+ - \mathsf{E}X_0\right)$$

Proof. Since two of the statements are similar, we only show first one, $\forall\, \lambda > 0$, define:

$$T = \inf\{t \geq 0 : X_t > \lambda\} \implies \mathsf{P}\left(\sup_{0\leq s\leq t} X_s > \lambda\right) = \mathsf{P}(T \leq t)$$

, then consider a stopped process, $\{X_{t\wedge T}, t \geq 0\}$, notice that:

$$\begin{aligned}\lambda\mathsf{P}\left(\sup_{0\leq s\leq t} X_s > \lambda\right) &= \lambda\mathsf{P}(T \leq t) = \lambda\mathsf{E}\mathbf{1}_{\{T\leq t\}} \leq \mathsf{E}X_T\mathbf{1}_{\{T\leq t\}} \\ &= \mathsf{E}X_{t\wedge T}\mathbf{1}_{\{T\leq t\}} = \mathsf{E}X^+_{t\wedge T}\mathbf{1}_{\{T\leq t\}} \leq \mathsf{E}X^+_{t\wedge T} \leq \mathsf{E}X^+_t\end{aligned}$$

Hence, we completed proof, by transfering original question to new stopping time. □

The above theorem is to consider the tail probability, and next one is for moment:

Theorem 2.1.9. *Let $X = \{X_t, t \geq 0\}$ be submartingale, then $\forall\, p > 1$:*

$$\mathsf{E}\left(\sup_{0\leq s\leq t} |X_s|\right)^p \leq \left(\frac{p}{p-1}\right)^p \mathsf{E}\,|X_t|^p$$

Proof. Similar to previous inequality, here we need more dedicate tail probability estimation, where we first define $Y = \sup_{0\leq s\leq t} |X_s|$, then:

$$\begin{aligned}\lambda\mathsf{P}\left(Y \geq \lambda\right) &\leq \mathsf{E}\,|X_{T\wedge t}|\,\mathbf{1}_{\{T\leq t\}} = \mathsf{E}\,|X_{T\wedge t}| - \mathsf{E}\,|X_{T\wedge t}|\,\mathbf{1}_{\{T>t\}} \\ &\leq \mathsf{E}\,|X_t| - \mathsf{E}\,|X_t|\,\mathbf{1}_{\{T>t\}} = \mathsf{E}\,|X_t| - \mathsf{E}\,|X_t|\,\mathbf{1}_{\{Y<\lambda\}} = \mathsf{E}\,|X_t|\,\mathbf{1}_{\{Y\geq\lambda\}}\end{aligned}$$

Then, using basic calculus to expand the moment:

$$\begin{aligned}\mathsf{E}Y^p &= \int_0^\infty p\lambda^{p-1}\mathsf{P}(Y \geq \lambda)\mathrm{d}\lambda \leq \int_0^\infty p\lambda^{p-2}\mathsf{E}\,|X_t|\,\mathbf{1}_{\{Y\geq\lambda\}}\mathrm{d}\lambda \\ &= \mathsf{E}\,|X_t| \int_0^\infty p\lambda^{p-2}\mathbf{1}_{\{Y\geq\lambda\}}\mathrm{d}\lambda = \mathsf{E}\,|X_t| \int_0^Y p\lambda^{p-2}\mathrm{d}\lambda \\ &= \frac{p}{p-1}\mathsf{E}\,|X_t|\,Y^{p-1} \leq \frac{p}{p-1}\left(\mathsf{E}\,|X_t|^p\right)^{1/p}\left(\mathsf{E}Y^p\right)^{(p-1)/p}\end{aligned}$$

The last inequality is due to holder's inequality, then we arrange terms to get:

$$\left(\mathsf{E}Y^p\right)^{1/p} \leq \frac{p}{p-1}\left(\mathsf{E}\,|X_t|^p\right)^{1/p}$$

For unbounded case, we stopped the process by $Y \wedge k$, and then sending k to inifinity, hence we could easily completed the proof. □

2.2 The Doob-Meyer Decomposition

Recall the discretised version of **Doob Decomposition**:

$$\underbrace{X_n}_{\text{Submartingale}} = \underbrace{M_n}_{\text{Martingale}} + \underbrace{A_n}_{\substack{\text{Predictable}\\ \text{Increasing}}}$$

The intuition is simple, from above construction:

$$\mathsf{E}\left(M_{n+1} \mid \mathcal{F}_n\right) = M_n \implies \mathsf{E}\left(X_{n+1} - A_{n+1} \mid \mathcal{F}_n\right) = X_n - A_n$$

Using the predictability of $\{A_n, n \in \mathbb{N}\}$, i.e. $A_n \in \mathcal{F}_{n-1}$:

$$\mathsf{E}\left(X_{n+1} \mid \mathcal{F}_n\right) - A_{n+1} = X_n + A_n \implies A_{n+1} - A_n = \mathsf{E}\left(X_{n+1} \mid \mathcal{F}_n\right) - X_n \geq 0$$

Since $\{X_n, n \in \mathbb{N}\}$ is submartingale, then in general, it is increasing, hence A_n is to capture the additional increasing parts, and martingale can be seen as "white noise", by iterating, and by default to set $A_0 = 0$, we have:

$$A_n = \sum_{k=0}^{n-1} \left[\mathsf{E}\left(X_{k+1} \mid \mathcal{F}_k\right) - X_k\right]$$

Hence, this subsection is to delve into such decomposition with continuous setting:

$$\underbrace{X_t}_{\text{Submartingale}} = \underbrace{M_t}_{\text{Martingale}} + \underbrace{A_t}_{\substack{\text{Predictable} \\ \text{Increasing}}}$$

, where assuming $\{X_t, t \geq 0\}$ is **right-continuous** submartingale. First is to consider increasing process in continuous setting:

Definition 2.2.1. An adapted process $A = \{A_t, t \geq 0\}$ is called **increasing** if:

(a) $A_0 = 0$ a.s.

(b) $t \mapsto A_t$ is non-decreasing and right-continuous for ω-a.e.

(c) $\mathsf{E}A_t < \infty$, $\forall\, t \geq 0$.

Remark 2.2.1. Note that for integrability condition is a little different from (c):

$$\mathsf{E}A_\infty = \mathsf{E}\lim_{t\to\infty} A_t < \infty$$

Besides increasing, other than predictable, we introduce natural:

Definition 2.2.2. An increasing process $A = \{A_t, t \geq 0\}$ is called **natural** if $\forall$ bounded, right continuous martingale, $M = \{M_t, t \geq 0\}$, then $\forall\, t \geq 0$:

$$\mathsf{E}\int_0^t M_s \mathrm{d}A_s = \mathsf{E}\int_0^t M_{s-} \mathrm{d}A_s$$

Remark 2.2.2. It is easy to see , if $s \mapsto A_s$ is a.s. continuous, then A is natural, since $M_s - M_{s-} \neq 0$ only for jump points, for a.s. continuity, there are only finite of them.

It seems wired to propose this definition, by thinking from discretised version:

$$\begin{aligned} A_n \in \mathcal{F}_{n-1} &\iff \text{predictable} \\ A_t \in \mathcal{F}_{t-} &\iff \text{"predictable"} \end{aligned}$$

Hence, we first check its discrete-time version:

Definition 2.2.3. An adapted and increasing sequence, $A = \{A_n, n \in \mathbb{N}\}$ is called **natural** if $\forall$ bounded martingale, $M = \{M_n, n \in \mathbb{N}\}$, then, $\forall\, n \in \mathbb{N}$:

$$\mathsf{E}\left(M_n A_n\right) = \mathsf{E}\left[\sum_{k=1}^{n} M_{k-1}\left(A_k - A_{k-1}\right)\right]$$

Then we are eager to see the connection between **natural** and **predictable**:

Proposition 2.2.1. *For in creasing sequence, A is predictable iff it is natural.*

Proof. The proof is direct, mainly by dedicate algebra:

$\Longrightarrow$ To prove, it is natural, it is same to show for every martingale M:

$$\mathsf{E}\left(M_n A_n\right) = \mathsf{E}\left[\sum_{k=1}^{n} M_{k-1}\left(A_k - A_{k-1}\right)\right] = \sum_{k=1}^{n}\left[\mathsf{E}M_{k-1}A_k - \mathsf{E}M_{k-1}A_{k-1}\right]$$

Now we could rearrange things to put minus term to left hand side:

$$\sum_{k=1}^{n} \mathsf{E}M_{k-1}A_{k-1} + \mathsf{E}M_nA_n = \sum_{k=1}^{n} \mathsf{E}M_{k-1}A_k$$

Since $A_0 = 0$ and M_nA_n serves as final term of first summation, then:

$$\sum_{k=1}^{n} \mathsf{E}M_kA_k = \sum_{k=1}^{n} \mathsf{E}M_{k-1}A_{k-1} \implies \sum_{k=1}^{n} \mathsf{E}\left(M_k - M_{k-1}\right)A_k = 0$$

From this equivalent statement, we plug in the predictability, if $A_n \in \mathcal{F}_{n-1}$:

$$\mathsf{E}A_n(M_n - M_{n-1}) = \mathsf{E}\left[A_n\mathsf{E}\left(M_n - M_{n-1} \mid \mathcal{F}_{n-1}\right)\right] = 0$$

Then each term of the summation is zero, we completed this direction.

$\Longleftarrow$ Assume A is natural, and $\forall$ bounded martingale M, from abovem, then:

$$\mathsf{E}A_n\left(M_n - M_{n-1}\right) = 0,\ \forall\, n \geq 1$$

Then goal is to show predictability, i.e. $A_n \in \mathcal{F}_{n-1}$, or $A_n = \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)$ a.s., now since A is natural, then for every bounded M, we notice that:

$$\begin{aligned}
&\mathsf{E}M_n\left[A_n - \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\right] = \mathsf{E}A_nM_{n-1} - \mathsf{E}\left[M_n\mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\right] \\
&= \mathsf{E}M_{n-1}\left[A_n - \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\right] + \mathsf{E}\left[\mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\left(M_n - M_{n-1}\right)\right] \\
&= \underbrace{\mathsf{E}M_{n-1}A_n - \mathsf{E}\mathsf{E}\left(A_nM_{n-1} \mid \mathcal{F}_{n-1}\right)}_{=0} + \mathsf{E}\left[\mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\underbrace{\mathsf{E}\left(M_n - M_{n-1} \mid \mathcal{F}_{n-1}\right)}_{=0}\right] = 0
\end{aligned}$$

Hence, by the arbitrarity, for each fixed n, we choose:

$$\xi_n = \operatorname{sgn}\left(A_n - \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\right)$$

, and define our martingale as:

$$M_k = \begin{cases} \mathsf{E}\left(\xi_n \mid \mathcal{F}_k\right) & 0 \le k \le n-1 \\ \xi_n & k \ge n \end{cases}$$

, which is bounded, and martingle, from this well-designed martingale, we know:

$$0 = \mathsf{E} M_n \left[A_n - \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\right] = \mathsf{E}\left|A_n - \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right)\right|$$

, above gives us:

$$A_n = \mathsf{E}\left(A_n \mid \mathcal{F}_{n-1}\right) \text{ a.s.} \quad \Longrightarrow \quad A_n \in \mathcal{F}_{n-1}$$

Hence, we completed the proof by showing both directions. □

Then next thing is to bridge the discrete one with continuous one:

$$\mathsf{E}\left(M_n A_n\right) = \mathsf{E}\left[\sum_{k=1}^{n} M_{k-1}\left(A_k - A_{k-1}\right)\right] \Longleftrightarrow \mathsf{E}\int_0^t M_s \mathrm{d}A_s = \mathsf{E}\int_0^t M_{s-} \mathrm{d}A_s$$

Hence, use the traditional discretisation method, by partition, $\Pi = \{t_0, t_1, \cdots, t_n\} \subseteq [0, t]$ with $0 = t_0 < t_1 < t_2 < \cdots < t_n = t$ and by setting $\|\Pi\| = \max_{1 \le k \le n} |t_k - t_{k-1}|$:

$$\int_0^t M_s \mathrm{d}A_s = \lim_{\|\Pi\| \to 0} \int_0^t M_s^{\Pi} \mathrm{d}A_s$$

, where by right continuous, we chose our discrete martingale to be:

$$M_s^{\Pi} = \sum_{k=1}^{n} M_{t_k} \mathbf{1}_{(t_{k-1}, t_k]}(s)$$

Then, by this setup, for each Π, we could see:

$$\begin{aligned} \mathsf{E}\int_0^t M_s^{\Pi} \mathrm{d}A_s &= \mathsf{E}\left[\sum_{k=1}^{n} M_{t_k}\left(A_{t_k} - A_{t_{k-1}}\right)\right] = \mathsf{E} M_{t_n} A_{t_n} + \sum_{k=1}^{n-1} \mathsf{E} M_{t_k} A_{t_k} - \sum_{k=2}^{n} \mathsf{E} M_{t_k} A_{t_{k-1}} \\ &= \mathsf{E} M_t A_t + \sum_{k=1}^{n-1} \mathsf{E} M_{t_k} A_{t_k} - \sum_{k=1}^{n-1} \mathsf{E} M_{t_{k+1}} A_{t_k} \\ &= \mathsf{E} M_t A_t - \sum_{k=1}^{n-1} \mathsf{E}\left[A_{t_k} \underbrace{\mathsf{E}\left(M_{t_{k+1}} - M_{t_k} \mid \mathcal{F}_{t_k}\right)}_{=0}\right] = \mathsf{E} M_t A_t \end{aligned}$$

Hence, we see the connection between discrete version and continuous one, next:

> **Definition 2.2.4.** Let $S = \{$ stopping times T with $T < \infty$ a.s. $\}$, we say a right-continuous $X_t, t \ge 0$ is of **class D** if $\{X_T, T \in S\}$ is u.i., furthermore, to fix $a > 0$, let $S_a = \{$ stopping times T with $Tlea$ a.s. $\}$, then $\{X_t, t \ge 0\}$ is of **class DL** if $\{X_T, T \in S_a\}$ is u.i. for every $a > 0$.

Then Doobs and Meyer found membership of class DL is sufficient for decomposition:

Theorem 2.2.2. *(Doob-Meyer Decomposition)*
If $\{X_t, t \geq 0\}$ is of ***class DL***, *and a submartingale, then it admits, $\forall\, t \geq 0$:*

$$X_t = M_t + A_t$$

, with $\{M_t, t \geq 0\}$ is martingale, $\{A_t, t \geq 0\}$ is increasing process. In particular, if we resrtict $\{A_t, t \geq 0\}$ to be natural, then such decomposition is unique.

Proof. Let us only consider the special cases, where we admit A is natural:

(1) For uniqueness, we assume it admits two different decomposition:

$$X_t = \begin{cases} M_t + A_t \\ M_t' + A_t' \end{cases} \implies B_t := M_t - M_t' = A_t' - A_t$$

, where $\{B_t, t \geq 0\}$ is martingale and of bounded variation , also natural, from the definition and exercise before, $\forall$ bounded continuous martingale $\{\xi_t, t \geq 0\}$:

$$\mathsf{E}\xi_t(A_t - A_t') = \mathsf{E}\int_0^t \xi_{s-}(\mathrm{d}A_s - \mathrm{d}A_s') = \mathsf{E}\int_0^t \xi_{t-}\mathrm{d}B_s = \lim_{n\to\infty}\mathsf{E}\sum_{j=1}^{m_n}\xi_{t_{j-1}^{(n)}}\left(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}\right)$$

, where $\Pi_n = \{t_0^{(n)}, t_1^{(n)}, \cdots, t_{m_n}^{(n)}\} \subseteq [0, t]$, for each term:

$$\mathsf{E}\xi_{t_{t_{j-1}}^{(n)}}\left(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}}\right) = \mathsf{E}\left[\xi_{t_{t_{j-1}}^{(n)}}\underbrace{\mathsf{E}\left(B_{t_j^{(n)}} - B_{t_{j-1}^{(n)}} \mid \mathcal{F}_{t_{j-1}^{(n)}}\right)}_{=0}\right] = 0$$

, then by similar trick, we obtained:

$$\mathsf{E}\,|A_t - A_t'| = 0 \implies A_t = A_t' \text{ a.s. } \forall\, t > 0 \implies A_t = A_t' \text{ a.s. } \forall\, t \in \mathbb{Q}^+$$

Lastly, by right-continuity, we achieved $A_t = A_t'$ a.s. $\forall\, t \geq 0$.

(2) For existence, fix $a > 0$, it suffices to construct the decomposition on $[0, a]$, consider:

$$Y_t := X_t - \mathsf{E}(X_a \mid \mathcal{F}_t) \leq 0$$

Then $\{Y_t, 0 \leq t \leq a\}$ is right-continuous submartingale.Consider dyadic partition:

$$\Pi_n = \left\{t_0^{(n)}, t_1^{(n)}, \cdots, t_{2^n}^{(n)}\right\}, \quad t_j^{(n)} = \frac{j}{2^n}a, \quad 0 \leq j \leq 2^n$$

For each $n \geq 1$, the discrete-time submartingale admits the Doob decomposition

$$Y_{t_j^{(n)}} = M_{t_j^{(n)}}^{(n)} + A_{t_j^{(n)}}^{(n)}, \qquad 0 \leq j \leq 2^n$$

where $A_0^{(n)} = 0$, and for $1 \leq j \leq 2^n$,

$$A_{t_j^{(n)}}^{(n)} = \sum_{k=0}^{j-1}\mathsf{E}\left(Y_{t_{k+1}^{(n)}} - Y_{t_k^{(n)}} \,\middle|\, \mathcal{F}_{t_k^{(n)}}\right)$$

In particular, $\{A^{(n)}_{t^{(n)}_j}\}_{0\le j\le 2^n}$ is predictable and increasing, notice:

$$Y_a = X_a - \mathsf{E}(X_a \mid \mathcal{F}_a) = 0 \implies 0 = Y_a = M^{(n)}_a + A^{(n)}_a$$

Hence, we may rewrite this into:

$$M^{(n)}_{t^{(n)}_j} = \mathsf{E}\big(M^{(n)}_a \mid \mathcal{F}_{t^{(n)}_j}\big) = -\mathsf{E}\big(A^{(n)}_a \mid \mathcal{F}_{t^{(n)}_j}\big) \implies Y_{t^{(n)}_j} = A^{(n)}_{t^{(n)}_j} - \mathsf{E}\big(A^{(n)}_a \mid \mathcal{F}_{t^{(n)}_j}\big)$$

We now claim that the family $\{A^{(n)}_a\}_{n\ge1}$ is u.i.(you may verifty). Hence, by the Dunford-Pettis compactness criterion, there exist an integrable random variable A_a and a subsequence $\{A^{(n_k)}_a\}_{k\ge1}$ such that $A^{(n_k)}_a \xrightarrow{L^1} A_a$, i.e.

$$\lim_{k\to\infty} \mathsf{E}(\xi A^{(n_k)}_a) = \mathsf{E}(\xi A_a)$$

for every bounded random variable ξ. To simplify notation, we relabel this subsequence again by $\{A^{(n)}_a\}$, now define

$$A_t := Y_t + \mathsf{E}(A_a \mid \mathcal{F}_t), \qquad 0 \le t \le a$$

where we choose its right-continuous modification. Likewise, for each n, define

$$A^{(n)}_t := Y_t + \mathsf{E}(A^{(n)}_a \mid \mathcal{F}_t), \qquad 0 \le t \le a$$

again with the right-continuous version. Then, for every $t \in \Pi_n$, this agrees with the discrete compensator constructed above, let

$$\mathbb{T} := \bigcup_{n=1}^{\infty} \Pi_n$$

By the weak L^1 stability of conditional expectation, for every $t \in \mathbb{T}$ and every bounded random variable ξ, we have

$$\lim_{n\to\infty} \mathsf{E}\big(\xi A^{(n)}_t\big) = \mathsf{E}(\xi A_t)$$

Now take $s, t \in \mathbb{T}$ with $0 \le s < t \le a$, and let ξ be bounded and nonnegative. For all large n, both s, t belong to Π_n, hence

$$\mathsf{E}\big[\xi(A_t - A_s)\big] = \lim_{n\to\infty} \mathsf{E}\big[\xi(A^{(n)}_t - A^{(n)}_s)\big] \ge 0$$

because $\{A^{(n)}_u\}_{u\in\Pi_n}$ is increasing. Choosing $\xi = \mathbf{1}_{\{A_s > A_t\}}$, we obtain

$$A_s \le A_t \quad \text{a.s.}$$

Since $\mathbb{T}$ is countable and by right-continuity, it is then nondecreasing on the whole interval $[0, a]$. Therefore, $\{A_t, 0 \le t \le a\}$ is an increasing process. Next:

$$Y_t = A_t - \mathsf{E}(A_a \mid \mathcal{F}_t) = X_t - \mathsf{E}(X_a \mid \mathcal{F}_t) \implies X_t = A_t + \mathsf{E}(X_a - A_a \mid \mathcal{F}_t)$$

Define $M_t := \mathsf{E}(X_a - A_a \mid \mathcal{F}_t)$, then it is a right-continuous martingale, and

$$X_t = M_t + A_t$$

It remains to show that A is natural. Let $\{\xi_t, 0 \le t \le a\}$ be any bounded right-continuous martingale. For each n, since $\{A^{(n)}_{t^{(n)}_j}\}_{0\le j\le 2^n}$ is predictable on Π_n:

$$\mathsf{E}(\xi_a A^{(n)}_a) = \mathsf{E}\left[\sum_{j=1}^{2^n} \xi_{t^{(n)}_{j-1}} \left(A^{(n)}_{t^{(n)}_j} - A^{(n)}_{t^{(n)}_{j-1}}\right)\right] = \mathsf{E}\left[\sum_{j=1}^{2^n} \xi_{t^{(n)}_{j-1}} \left(Y_{t^{(n)}_j} - Y_{t^{(n)}_{j-1}}\right)\right]$$
$$= \mathsf{E}\left[\sum_{j=1}^{2^n} \xi_{t^{(n)}_{j-1}} \left(A_{t^{(n)}_j} - A_{t^{(n)}_{j-1}}\right)\right]$$

Letting $n \to \infty$, the left-hand side converges to $\mathsf{E}(\xi_a A_a)$ by weak convergence:

$$\mathsf{E}(\xi_a A_a) = \mathsf{E}\int_0^a \xi_s \mathrm{d}A_s = \mathsf{E}\int_0^a \xi_{s-} \mathrm{d}A_s$$

, so A is natural. Hence, on $[0, a]$, we have constructed

$$X_t = M_t + A_t, \qquad 0 \le t \le a,$$

, with M a right-continuous martingale, and A a natural increasing process.

Then we completed this long proof. □

2.3 Square-Integrable Martingale

After a complicated derivation about Doob-Meyer decomposition, we had to know what it can do, then it comes to our approach. Here firstly propose a definition:

Definition 2.3.1. Let $X = \{X_t, t \ge 0\}$ be a **right-continuous** martingale, if:

(a) X is **square-integrable**, i.e.:

$$\mathsf{E}X_t^2 < \infty, \forall\, t \ge 0$$

(b) $X_0 = 0$

, then we say $X \in \mathcal{M}_2$.

Notice that, if $X \in \mathcal{M}^2$, then $Y := X^2 := \{X_t^2, t \ge 0\}$ is a submartingale, to make it meaningful, we require it to be integrable, which explains why we are interested into square-integrable. Since it is square integrable, then it is clearly in DL, then it admits Doob-Meyer decomposition:

$$Y_t = M_t + A_t$$

Furthermore, if we restrict all the sample path of X to be continuous, noted as $X \in \mathcal{M}_2^c$, then from last subsection, M_t and A_t be continuous. Empowered by this, we define:

Definition 2.3.2. $\forall\, X \in \mathcal{M}_2$, the **quadratic variation** of X is defined by:

$$\langle X\rangle_t := A_t$$

Remark 2.3.1. By theorem **??**, $\langle X\rangle_t$ is the unique, adapted, natural, increasing process, such that $\langle X\rangle_0 = 0$, moreover, $X_t^2 - \langle X\rangle_t$ is martingale.
Now, if we take, $X, Y \in \mathcal{M}_2$, then:

$$\begin{cases} (X_t + Y_t)^2 - \langle X+Y\rangle_t \\ (X_t - Y_t)^2 - \langle X-Y\rangle_t \end{cases} \quad \text{are both martingale.}$$

, then we could do substraction of these two:

$$4X_tY_t - \big(\langle X+Y\rangle_t - \langle X-Y\rangle_t\big) \text{ is also martingale.}$$

Inspired by this, we could define the **cross variation**:

Definition 2.3.3. For $X, Y \in \mathcal{M}_2$, we define the **cross variation**:

$$\langle X, Y\rangle_t := \frac{1}{4}\big(\langle X+Y\rangle_t - \langle X-Y\rangle_t\big)$$

, such that $X_tY_t - \langle X, Y\rangle_t$ is a martingale.

Notice that this definition recovers our quadratic variation, since:

$$\langle X, X\rangle_t = \frac{1}{4}\langle 2X\rangle = \frac{1}{4}\cdot 4\cdot A_t = \langle X\rangle_t$$

But more commonly, quadratic or generally, p-th variation is defined by:

Definition 2.3.4. The **p-th variation** of X over partition:

$$\Pi = \{t_0, t_1, \cdots, t_m\} \subseteq [0, t]$$

is defined to be:

$$V_t^{(p)}(\Pi) = \sum_{k=1}^{m} \left|X_{t_k} - X_{t_{k-1}}\right|^p$$

Hence, we want to justify, under class of $\mathcal{M}_2^c$, it is "equivalent":

Theorem 2.3.1. *Let $X \in \mathcal{M}_2^c$, for partition, Π, of $[0, t]$, we have:*

$$\lim_{\|\Pi\|\to 0} V_t^{(2)}(\Pi) = \langle X\rangle_t, \ \textit{in probability}$$

, i.e., $\forall\, \varepsilon > 0, \eta > 0$, $\exists\, \delta > 0$, s.t.:

$$\|\Pi\| < \delta \implies \mathsf{P}\left(\left|V_t^{(2)}(\Pi) - \langle X\rangle_t\right| > \varepsilon\right) < \eta$$

Proof. To see it is "equal" in probability, we consider L^2 distance:

$$\begin{aligned} \mathsf{E}\left(V_t^{(2)}(\Pi) - \langle X\rangle_t\right)^2 &= \mathsf{E}\left[\sum_{k=1}^{m}\left|X_{t_k} - X_{t_{k-1}}\right|^2 - \langle X\rangle_t\right]^2 \\ &= \mathsf{E}\left\{\sum_{k=1}^{m}\left[\left|X_{t_k} - X_{t_{k-1}}\right|^2 - \left(\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right)\right]\right\}^2 \end{aligned}$$

Hence we need to consider them seperately:

(1) Firstly, for centered terms, notice that, for $t_j < t_{i-1}$:

$$\mathsf{E}\left[\left(X_{t_i} - X_{t_{i-1}}\right)^2 \mid \mathcal{F}_{t_j}\right] = \mathsf{E}\left(X_{t_i}^2 \mid \mathcal{F}_{t_j}\right) - 2\mathsf{E}\left(X_{t_i}X_{t_{i-1}} \mid \mathcal{F}_{t_j}\right) + \mathsf{E}\left(X_{t_{i-1}}^2 \mid \mathcal{F}_{t_j}\right)$$

Furthermore, if we focus on cross term, we may stick in a conditional expectation:

$$\begin{aligned}\mathsf{E}\left(X_{t_i}X_{t_{i-1}} \mid \mathcal{F}_{t_j}\right) &= \mathsf{E}\left[\mathsf{E}\left(X_{t_i}X_{t_{i-1}} \mid \mathcal{F}_{t_{i-1}}\right) \mid \mathcal{F}_{t_j}\right] \\ &= \mathsf{E}\left[X_{t_{i-1}}\mathsf{E}\left(X_{t_i} \mid \mathcal{F}_{t_{i-1}}\right) \mid \mathcal{F}_{t_j}\right] = \mathsf{E}\left(X_{t_{i-1}}^2 \mid \mathcal{F}_{t_j}\right)\end{aligned}$$

Therefore, we could see the cross term is gone, since:

$$\mathsf{E}\left[\left(X_{t_k} - X_{t_{k-1}}\right)^2 \mid \mathcal{F}_{t_j}\right] = \mathsf{E}\left(X_{t_k}^2 - X_{t_{k-1}}^2 \mid \mathcal{F}_{t_j}\right) = \mathsf{E}\left(\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}} \mid \mathcal{F}_{t_j}\right)$$

By symmetry, another parts also works.

(2) Secondly, for centered term, we use inequlity to divide and conquer:

$$\begin{aligned}&\sum_{k=1}^{m} \mathsf{E}\left[\left|X_{t_k} - X_{t_{k-1}}\right|^2 - \left(\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right)\right]^2 \\ &\quad \leq \sum_{k=1}^{m}\left[2\mathsf{E}\left(X_{t_k} - X_{t_{k-1}}\right)^4 + 2\mathsf{E}\left(\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right)^2\right] \\ &\quad = 2\,\mathsf{E}V_t^4(\Pi) + 2\sum_{k=1}^{m}\mathsf{E}\left(\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right)^2 \\ &\quad \leq 2\,\mathsf{E}V_t^4(\Pi) + 2\,\mathsf{E}\left[\sup_{1\leq k\leq m}\left|\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right| \cdot \langle X\rangle_t\right]\end{aligned}$$

, where we only used a simple inequality, $(a-b)^2 \leq 2a^2 + 2b^2$. Then next step is to prove these two terms go to zero:

(i) For blued term, we provide a lemma without proof:

> **Lemma.** *Let $X \in \mathcal{M}_2^c$ satisfy $|X_s| \leq K < \infty$, $\forall\ s \leq t$, then:*
>
> $$\mathsf{E}\left[V_t^{(2)}(\Pi)\right]^2 \leq 6K^4$$

Assuming this, you should verify, we can see:

$$\mathsf{E}V_t^{(4)}(\Pi) \leq \mathsf{E}\left(\sum_{k=1}^{m}\left|X_{t_k} - X_{t_{k-1}}\right|^2 \cdot m_t\left(X, \|\Pi\|\right)\right)$$

, where $m_t(X,\delta)$ is, the δ-distance, defined by:

$$m_t(X,\delta) = \sup_{0\leq r<s\leq t}\left\{|X_r - X_s| : |s-r| \leq \delta\right\}$$

Then, under the above lemma condition, if $X \in \mathcal{M}_2^c$ and $|X_s| \leq K, \forall\, 0 \leq s \leq t$ by bounded convergence:

$$\mathsf{E} m_t\left(X, \|\Pi\|\right) \xrightarrow{\|\Pi\|\to 0} 0$$

Now, using Cauchy-Schwartz, to seperate and control them:

$$\mathsf{E} V_t^{(4)}(\Pi) \leq \left\{\mathsf{E}\left[V_t^{(2)}(\Pi)\right]^2\right\}^{1/2} \left[\mathsf{E} m_t^4\left(X, \|\Pi\|\right)\right]^{1/2} \leq \sqrt{6}K^2 \cdot \left[\mathsf{E} m_t^4\left(X, \|\Pi\|\right)\right]^{1/2} \to 0$$

(a) For greened term, also introduce new lemma omitted proof:

> **Lemma.** *Let $X \in \mathcal{M}_2^c$ satisfy $\langle X\rangle_s \leq K < \infty$, $\forall\, 0 \leq s \leq t$, then:*
>
> $$\mathsf{E}\left[\sup_{1\leq k\leq m} \left|\langle X\rangle_{t_k} - \langle X\rangle_{t_{k-1}}\right| \cdot \langle X\rangle_t\right] \longrightarrow 0$$

This can be done by uniform continuity.

Lastly, we need to deal with general $X \in \mathcal{M}_2^c$, i.e. unbounded, then we use the technique, "localisation", to define, $\forall\, n \geq 1$:

$$T_n = \inf\left\{t \geq 0 : |X_t| \geq n \text{ or } \langle X\rangle_t \geq n\right\}$$

Then, considered localised process, $X_t^{(n)} = X_{t\wedge T_n}$ is bounded martingale, also as the following squred process:

$$\left\{X_{t\wedge T_n}^2 - \langle X\rangle_{t\wedge T_n}, t \geq 0\right\}$$

, where it is done by the uniqueness of $\langle X^{(n)}\rangle_t$, then we get:

$$\langle X^{(n)}\rangle_t = \langle X\rangle_{t\wedge T_n}$$

For this localised process, from above discussion we know:

$$\mathsf{E}\left[\sum_{k=1}^m \left|X_{t_k\wedge T_n} - X_{t_{k-1}\wedge T_n}\right|^2 - \langle X\rangle_{t\wedge T_n}\right]^2 \xrightarrow{\|\Pi\|\to 0} 0$$

Unfornately, we cannot directly send $n \to \infty$, since there are several variables in control, however, which can be done by ε-δ language, now, $\forall\, \varepsilon > 0, \forall\, \eta > 0$, we can always choose a large enough fixed $t > 0$ s.t.

$$\mathsf{P}(T_n < t) \leq \frac{\eta}{2}$$

Besides, $\exists\, \delta > 0$, s.t. $\|\Pi\| \leq \delta$, here we can choose $\|\Pi\|$ small enough s.t.:

$$\mathsf{E}\left[\sum_{k=1}^m \left|X_{t_k\wedge T_n} - X_{t_{k-1}\wedge T_n}\right|^2 - \langle X\rangle_{t\wedge T_n}\right]^2 \leq \varepsilon^2 \cdot \frac{\eta}{2}$$

Then, by simple event seperation and Markov Inequality:

$$\begin{aligned}
\mathsf{P}\left(\left|V_t^{(2)}(\Pi) - \langle X\rangle_t\right| > \varepsilon\right) &= \mathsf{P}(T_n < t) + \mathsf{P}\left(\left|V_{t\wedge T_n}^{(2)}(\Pi) - \langle X\rangle_{t\wedge T_n}\right| > \varepsilon\right) \\
&\leq \frac{\eta}{2} + \frac{1}{\varepsilon}\mathsf{E}\left[\sum_{k=1}^m \left|X_{t_k\wedge T_n} - X_{t_{k-1}\wedge T_n}\right|^2 - \langle X\rangle_{t\wedge T_n}\right]^2 \leq \eta
\end{aligned}$$

Hence, we completed the proof by using ε-δ language to bypass. $\square$

Since the power 2 gives you an exact convergence, then if the power changes a little:

Corollary 2.3.2. *Same setting as before,* $\forall\, q > 2$*:*

$$V_t^{(q)} \xrightarrow{p.} 0$$

, moreover, on $\{\langle X \rangle_t \geq 0\}$*,* $\forall\, q < 2$*:*

$$V_t^{(q)} \xrightarrow{p.} \infty$$

The conclusion to be drawn from above is that for continuous, square-integrable martingales,quadratic variation is the "right" variation to study. All variations of higher order are zero, and except in trivial cases, where the martingale is a.s. constant. Thu, the sample paths of continuous, square-integrable martingales are quite different from "ordinary" continuous martingale. Being of unbounded first variation, they cannot be differentiable, nor is it possible to define integral of form:

$$\int_0^t Y_s \mathrm{d} X_s$$

, w.r.t $X \in \mathcal{M}_2^c$ in a pathwise Lebesgue-Stieltjes sense.

Chapter 3

Stochastic Integration

In the last section, we focus on continuous square-integrable martingale class:

$$\mathcal{M}_2 = \{X : \mathsf{E}X_t^2 < \infty, \forall\, t \geq 0\}$$

Before, we move into integration, we delve into this space, by restricting it first:

$$\mathcal{M}^2 := \left\{X : \sup_{t\geq 0} \mathsf{E}X_t^2 < \infty\right\}$$

, clearly this space is smaller, since it implies continous, square-integrable class.

3.1 The L^2 Theory of Martingales

Beside further, we will see it is a **Hilbert Space**, moreover it may be easy for us to study the process that starts at zero (otherwise, we may subtract the initial value):

$$\mathcal{M}_0^2 = \left\{X \in \mathcal{M}^2 : X_0 = 0\right\}$$

Theorem 3.1.1. *The space $\mathcal{M}^2$ is same as Hilbert space, $L^2(\mathcal{F}_\infty)$, with bijection:*

$$\{M_t, t \geq 0\} \in \mathcal{M}^2 \mapsto M_\infty := \lim_{t\to\infty} M_t \in L^2(\mathcal{F}_\infty)$$
$$M_\infty \in L^2(\mathcal{F}_\infty) \mapsto M_t := \mathsf{E}\left(M_\infty \mid \mathcal{F}_t\right) \in \mathcal{M}^2$$

Proof. omitted with intention. □

Remark 3.1.1. The live of space for backward is provided by:

$$\mathsf{E}X_t^2 = \mathsf{E}\left[\mathsf{E}\left(X_\infty \mid \mathcal{F}_t\right)^2\right] \leq \mathsf{E}\left[\mathsf{E}\left(X_\infty^2 \mid \mathcal{F}_t\right)\right] = \mathsf{E}X_\infty^2 < \infty$$

, then recall the Doob's L^2 inequality:

$$\mathsf{E}\sup_{0\leq s\leq t} |X_s|^2 \leq 4\mathsf{E}X_t^2$$

, sending $t \to \infty$ to see:

$$\mathsf{E}\sup_{t\geq 0} X_s^2 \leq 4\mathsf{E}X_\infty^2$$

Next interesting things about $\mathcal{M}^2$ space is that it is **complete**, i.e. for any cauchy squence $\{M^{(n)}\} \in \mathcal{M}^2$, we have $\{M_\infty^{(n)}\} \in L^2(\mathcal{F}_\infty)$ is also cauchy sequence:

Theorem 3.1.2. *The space $\mathcal{M}^2$ is complete.*

Proof. We only need to show, if $\exists\ M_\infty \in L^2(\mathcal{F}_\infty)$, s.t.:

$$M_\infty^{(n)} \xrightarrow{L^2} M_\infty$$

, with definition, $M_t = \mathsf{E}\left(M_\infty \mid \mathcal{F}_t\right)$, then:

$$\sup_{t\geq 0}\left|M_t^{(n)} - M_t\right| \xrightarrow{L^2} 0$$

, however, above is easy to show as we indicated in remark 3.1.1:

$$\begin{aligned}\mathsf{E}\sup_{t\geq 0}\left|M_t^{(n)} - M_t\right| &= \mathsf{E}\sup_{t\geq 0}\mathsf{E}\left(M_\infty^{(n)} - M_\infty \mid \mathcal{F}_t\right)^2 \\ &\leq 4\,\mathsf{E}\left(M_\infty^{(n)} - M_\infty\right)^2 \longrightarrow 0\end{aligned}$$

Therefore, we completed the proof. □

More interesting, we focus on the continuous part of this space:

Corollary 3.1.3. *The space, $c\mathcal{M}^2$, of continuous martingale in $\mathcal{M}^2$ is closed in the topology of $\mathcal{M}^2$.*

Proof. Notice, if $M^{(n)} \to M$, for $M^{(n)} \in c\mathcal{M}^2$, then:

$$\sup_{t\geq 0}\left|M_t^{(n)} - M_t\right| \xrightarrow{L^2} 0$$

, therefore, we could pick a subsequence, i.e. $\exists\ \{n_k, k \geq 0\}$, s.t.:

$$\sup_{t\geq 0}\left|M_t^{(n_k)} - M_t\right| \xrightarrow{a.s.} 0$$

, since each one is continuous, we reached M_t is continous. □

Then next thing is to think about cross variation:

Definition 3.1.1. For $X, Y \in \mathcal{M}^2$, we define:

$$\langle X, Y\rangle := \mathsf{E}X_\infty Y_\infty$$

To connect previous intuition, we define **weakly orthogonal**.

Definition 3.1.2. $\forall\ M, N \in \mathcal{M}_0^2$, we say M and N are **weakly orthogonal**, if:

$$\mathsf{E}M_\infty N_\infty = 0$$

Then correspondingly, we have **strongly orthogonal**:

Definition 3.1.3. $\forall\ M, N \in \mathcal{M}_0^2$, M and N are **strongly orthogonal**, if, $\forall$ s.t. T:

$$\mathsf{E} M_T N_T = 0$$

To better understand these orthogonality, we present following lemma:

Lemma 3.1.4. *M and N are strongly orthogonal iff MN is u.i.*

Proof. Here, we omit the proof, but introduce a new lemma for verifying u.i. □

Lemma 3.1.5. *For any progressively measurable process, s.t. $\forall$ s.t. T, if:*

(a) $\mathsf{E}\,|M_T| < \infty$;

(b) $\mathsf{E} M_T = 0$.

, then M is u.i. martingale.

As procedure in functional analysis or topology, we are interested into:

Definition 3.1.4. (Stable subspace of $\mathcal{M}_0^2$)
We say $\mathcal{N}_0 \subseteq \mathcal{M}_0^2$ is a stable subspace if:

(a) $\mathcal{N}_0$ is closed subspace;

(b) $\forall\ N \in \mathcal{N}_0$, $\forall$ s.t., T we have:

$$N^T \in \mathcal{N}_0$$

, where N^T should be understood as: $N_t^T = N_{t\wedge T}$.

Above definition sounds astounishing, here we present example:

Example 3.1.1. $c\mathcal{M}_0^2$ is stable subspace.

Empowered by this new tool, we may re-represent our old results:

Theorem 3.1.6. *Let $\mathcal{N}_0$ be a stable subspace of $\mathcal{M}_0^2$, let $\mathcal{N}_0^\perp$ be the set of $Z \in \mathcal{M}_0^2$, which are weakly orthogonal to every element of $\mathcal{N}_0$, i.e.:*

$$\mathcal{N}_0^\perp = \left\{ Z \in \mathcal{M}_0^2 : \langle Z, N \rangle = 0, \forall\ N \in \mathcal{N}_0 \right\}$$

, then $\mathcal{N}_0^\perp$ is a stable subspace of $\mathcal{M}_0^2$, and $\forall\ Z \in \mathcal{N}_0^\perp$, we have Z and N are strongly orthogonal for all $N \in \mathcal{N}_0$, finally, $\forall\ M \in \mathcal{M}_0^2$, it admits:

$$M = N + Z$$

Proof. We only show it is stable, the decomposition part is just Doob-Meyer's result.

(a) Firstly, it is easy to see, if $Z_n \xrightarrow{L^1} Z$, then:

$$\mathsf{E}Z_n N = 0 \implies \mathsf{E}ZN = 0$$

(b) Next, $\forall\ Z \in \mathcal{N}_0^\perp$, then $\forall\ N \in \mathcal{N}_0$:

$$\mathsf{E}Z_\infty N_\infty = 0$$

, we are left to show, $\forall$ s.t. T, we have $Z^T \in \mathcal{N}_0^\perp$, which is due to:

$$\begin{aligned}\mathsf{E}Z_\infty^T N_\infty &= \mathsf{E}Z_T N_\infty = \mathsf{E}\left[Z_T \mathsf{E}\left(N_\infty \mid \mathcal{F}_T\right)\right] = \mathsf{E}Z_T N_T \\ &= \mathsf{E}\left[\mathsf{E}\left(Z_\infty \mid \mathcal{F}_T\right) N_T\right] = \mathsf{E}Z_\infty N_T = \mathsf{E}Z_\infty N_\infty^T = 0\end{aligned}$$

, last step is by $N^T \in \mathcal{N}_0$.

Hence, it is stable, we completed the proof. □

Lastly, we consider the discontinuous part, or the complement of continuous:

Definition 3.1.5. Define the discountinuous space as:

$$d\mathcal{M}_0^2 = \left(c\mathcal{M}_0^2\right)^\perp$$

Then $\forall\ M \in \mathcal{M}_0^2$, we write:

$$M = M^c + M^d$$

, with $M^c \in c\mathcal{M}_0^2$ and $M^d \in d\mathcal{M}_0^2$ and $M^c \cdot M^d$ is u.i, by $\mathsf{E}M_T^c M_T^d = 0$, and jump part:

Definition 3.1.6. Define the jump as:

$$\Delta f(t) = f(t) - f(t-)$$

And Meyer develop following theorem to understand this by new representation:

Theorem 3.1.7. *Let $M \in \mathcal{M}_0^2$, then $\exists$ unique increasing process, $[M]$, null at 0, s.t.:*

(a) $M^2 - [M]$ is u.i.;

(b) $\Delta[M] = (\Delta M)^2$.

Proof. Omitted with intention. □

Remark 3.1.2. Intuitively, it is to say the quadratic variation process jumps at the same point as based-off martingale, and the intensity is its squared.

As similarly as before, for $M, N \in \mathcal{M}_0^2$, let:

$$[M, N] := \frac{1}{4}\left([M + N] - [M - N]\right)$$

Then Kunita and Watanabe studied the jump of this related to based-off martingale:

Theorem 3.1.8. *The process $[M, N]$ is the unique FV process null at 0 s.t.:*

(a) $MN - [M, N]$ *is u.i.;*

(b) $\Delta[M, N] = \Delta M \Delta N$.

Proof. Omitted with intention. $\square$

Remark 3.1.3. Intuitively, it is to say the cross variation jump iff two processes both jump.

3.2 Elementary Previsible Integrands

After those preparation, we restrict ourself into this class, let, $M \in \mathcal{M}_0^2$, now for two s.t.s $S \leq T$, and some bounded $Z \in \mathcal{F}_S$, if $H_t = Z\mathbf{1}_{(S,T]}(t)$, then we could define :

$$\int_0^t H_s \mathrm{d}M_s := (H \cdot M)(t) = Z(M_T - M_S)$$

$$\left\{= \int_0^t \mathbf{1}_{(a,b]} \mathrm{d}F(s) = F(t \wedge b) - F(t \wedge a)\right\}$$

As we discussion at motivation, we want to show $(H \cdot M) \in \mathcal{M}_0^2$:

(1) Firstly we will prove $\forall$ s.t. R:

(a) $\mathsf{E}\left|(H \cdot M)_R\right| < \infty$:

Proof. Let $K > 0$, s.t. $|Z| \leq K < \infty$, then:

$$\mathsf{E}\left|(H \cdot M)_R\right| \leq K\left[\mathsf{E}\left|M_{T \wedge R}\right| + \mathsf{E}\left|M_{S \wedge R}\right|\right] \leq 2K\mathsf{E}\left|M_\infty\right| < \infty$$

, where it is due to OST:

$$\mathsf{E}\left|M_{T \wedge R}\right| = \mathsf{E}\left[\left|\mathsf{E}\left(M_\infty \mid \mathcal{F}_{T \wedge R}\right)\right|\right] \leq \mathsf{E}\left[\mathsf{E}\left(\left|M_\infty\right| \mid \mathcal{F}_{T \wedge R}\right)\right] = \mathsf{E}\left|M_\infty\right|$$

Therefore, it is simply finished. $\square$

(b) $\mathsf{E}(H \cdot M)_R = 0$:

Proof. To prove above, it is equivalent to show:

$$\mathsf{E}Z\left(M_{T \wedge R} - M_{S \wedge R}\right) = 0$$

Considering replace it with indicator function since it is the basis of this space:

$$\text{For } A \in \mathcal{F}_S, \text{ if } Z = \mathbf{1}_A \implies \mathsf{E}\mathbf{1}_A\left(M_{T \wedge R} - M_{S \wedge R}\right) = \mathsf{E}\mathbf{1}_A\left(M_{T_A \wedge R} - M_{S_A \wedge R}\right) = 0$$

, where we could define:

$$S_A := \begin{cases} S, & \text{on } A \\ \infty, & \text{on } A^c \end{cases}$$

Then we completed the proof by specifying the indicator function. $\square$

Then it is to say $(H \cdot M)$ is u.i. martingale.

(2) Lastly, we remain to show it is L^2-bounded, which is easy to see, assume $|Z| \le K$:

$$\mathsf{E}\,(H \cdot M)_t^2 = \mathsf{E}\left[Z^2\big(M_{t\wedge T} - M_{t\wedge S}\big)^2\right] \le 2K^2 \mathsf{E} M_{t\wedge T}^2 + 2K^2 \mathsf{E} M_{t\wedge S}^2$$

Since, $M \in \mathcal{M}_0^2$, then M^2 is submartingale, by OST:

$$\mathsf{E}\,(H \cdot M)_t^2 \le 4K^2 \mathsf{E} M_t^2 \le 4K^2 \sup_{u\ge 0} \mathsf{E} M_u^2 < \infty \implies \sup_{t\ge 0} \mathsf{E}\,(H \cdot M)_t^2 < \infty$$

Then we showed $(H \cdot M) \in \mathcal{M}_0^2$, to extend this result, we have following lemma:

Lemma 3.2.1. *Let $S \le T$ be two s.t., if $Z \in b\mathcal{F}_S$ and M is u.i. martingale, define:*

$$H_t = Z\mathbf{1}_{(S,T]}(t)$$

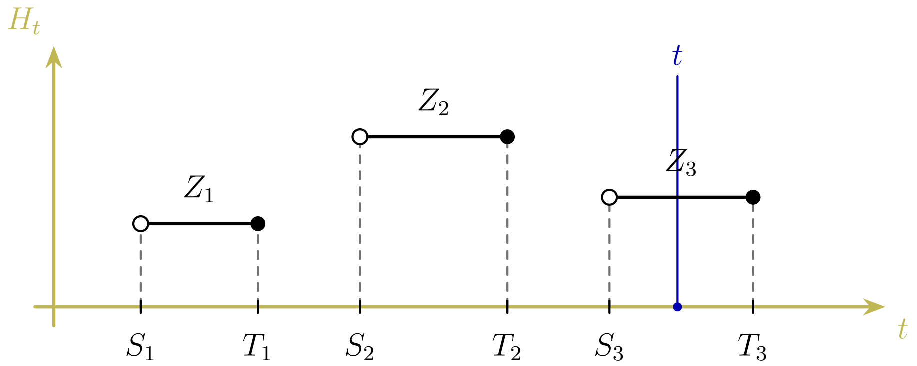
, then $(H \cdot M)$ is u.i. martingale.

Proof. Follow from above discussion. □

Next, a natural idea is to extend our result to linear combination, considering:

$$H_t = \sum_{i=1}^{n} Z_i \mathbf{1}_{(S_i,T_i]}(t)$$

, where each $Z_i \in b\mathcal{F}_{S_i}$, and a series s.t.s, s.t.:$S_1 \le T_1 \le S_2 \cdots$ as illustration below:



The left endpoint is not included, so Z_i is already known at time S_i.

Corollary 3.2.2. *Let $\{S_i, i \in [n]\}$, $\{T_i, i \in [n]\}$ be two series of s.t.s, s.t:*

$$S_i \le T_i, \ i \in [n]$$

, if $Z_i \in b\mathcal{F}_{S_i}$ and M is u.i. martingale, define:

$$H_t = \sum_{i=1}^{n} Z_i \mathbf{1}_{(S_i,T_i]}(t)$$

, then $(H \cdot M)$ is also u.i. martingale.

Proof. Directly follow from lemma 3.2.1. □

Therefore, a follow-up question is that, what is the largest class that inherit such property:

Definition 3.2.1. A previsible σ-algebra $\mathcal{P}$ on $(0,\infty)\times\Omega$ is defined to be the smallest σ-algebra s.t. every adapted L-process is $\mathcal{P}$-measurable.

Remark 3.2.1. The so called **L-process** is just what we used before, the L.C.R.L process, and correspondingly, we say $\{X_t, t\geq 0\}$ is previsible if it is $\mathcal{P}$-measurable.

Lemma 3.2.3. *Suppose S and T are two s.t.s with $S\leq T<\infty$, let $Z\in b\mathcal{F}_S$ and:*

$$H = Z\mathbf{1}_{(S,T]}$$

, then H defined above is previsible.

Proof. Here, $\forall\, B\in\mathcal{B}(\mathbb{R})$, and we assume $0\notin B$, then $\forall\, t>0$:

$$\begin{aligned}\{H_t\in B\} &= \{Z\in B\}\cap\{S<t\leq T\}\\ &= \{Z\in B\}\cap\{S<t\}\cap\{t\leq T\}\\ &= \bigcup_{n=1}^{\infty}\left(\underbrace{\{Z\in B\}\cap\{S\leq t-1/n\}}_{\in\mathcal{F}_{t-1/n}\subseteq\mathcal{F}_t}\right)\bigcap\underbrace{\{T\geq t\}}_{=\{T<t\}^c\in\mathcal{F}_t}\ \in\mathcal{F}_t\end{aligned}$$

Hence, we completed the proof. $\square$

Now, we want to see the connection of simple process and previsible process:

Definition 3.2.2. Define the σ-algebra of bounded simple process:

$$b\mathcal{E} = \left\{\sum_{i=1}^{n} Z_i\mathbf{1}_{(S_i,T_i]} : n\geq 1, Z_i\in b\mathcal{F}_{S_i}, S_1\leq T_1\leq S_2\leq\cdots\right\}$$

And then we claim that there are the same after completion:

Claim 3.2.4. $\sigma\left(b\mathcal{E}\right)=\mathcal{P}$

Proof. The forward is simple, and backward conclusion can be donw by using that fact:

$$H_t = \lim_{k\uparrow\infty}\lim_{n\uparrow\infty}\sum_{i=2}^{nk} H_{\frac{i-1}{n}}\mathbf{1}_{\left(\frac{i-1}{n},\frac{i}{n}\right]}(t)$$

We leave the details to the reader. $\square$

3.3 The L^2-Itô Integral

After discussion with integrands, we now relook at the L^2 theory:

$$\mathcal{M}_0^2\longleftrightarrow L^2\left(\mathcal{F}_\infty\right)$$

In the Itô integral, the integrand is allowed to remember the past, but not to foresee the future. This non-anticipative nature is exactly what predictability formalises.

Lemma 3.3.1. *Let $H \in b\mathcal{E}$, and $M \in \mathcal{M}_0^2$, then:*

$$(H \cdot M) \in \mathcal{M}_0^2$$

, and more specific, if $H = \sum_{i=1}^{n} Z_{i-1}\mathbf{1}_{(T_{i-1},T_i]}$, then:

$$\mathsf{E}\,(H \cdot M)_\infty^2 = \sum_{i=1}^{n} \mathsf{E}\left[Z_{i-1}^2\left(M_{T_i} - M_{T_{i-1}}\right)^2\right]$$

Proof. The first result have been showed in the preparation of section 3.2, now since:

$$(H \cdot M) \text{ is u.i. martingale } \implies (H \cdot M)_t \xrightarrow{L^1} (H \cdot M)_\infty := \sum_{i=1}^{n} Z_{i-1}\Delta_i$$

, where we denote, $\Delta_i = M_{T_i} - M_{T_{i-1}}$, now notice that:

$$\mathsf{E}\,(H \cdot M)_\infty^2 = \mathsf{E}\left[\sum_{i=1}^{n} Z_{i-1}^2\Delta_i^2\right] + 2\mathsf{E}\left[\sum_{i<j} Z_{i-1}\Delta_i Z_{j-1}\Delta_j\right]$$

, then it remains to show the cross terms is vanished, obviously, for $i \le j-1 < j$:

$$\mathsf{E}\left[Z_{i-1}\Delta_i Z_{j-1}\Delta_j\right] = \mathsf{E}\left[Z_{i-1}\Delta_i \cdot Z_{j-1}\mathsf{E}\left(\Delta_j \mid \mathcal{F}_{T_{j-1}}\right)\right] = 0$$

Lastly, by Doob's inequality, we also have:

$$\mathsf{E}\sup_{t\ge 0}\,(H \cdot M)_t^2 \le 4\mathsf{E}\,(H \cdot M)_\infty^2 < \infty$$

, hence $(H \cdot M) \in \mathcal{M}_0^2$, and we completed the proof. □

Next, recall that, if $M \in \mathcal{M}_0^2$, then $\exists$ quadratic variation, $[M]$, s.t.:

$$M^2 - [M] \text{ is u.i. martingale.}$$

Then, using this fact, notice that:

$$\begin{aligned}
\mathsf{E}\left[\left(M_{T_i} - M_{T_{i-1}}\right)^2 \mid \mathcal{F}_{T_{i-1}}\right] &= \mathsf{E}\left[M_{T_i}^2 \mid \mathcal{F}_{T_{i-1}}\right] + M_{T_{i-1}}^2 - 2\mathsf{E}\left[M_{T_i}M_{T_{i-1}} \mid \mathcal{F}_{T_{i-1}}\right] \\
&= \mathsf{E}\left[M_{T_i}^2 \mid \mathcal{F}_{T_{i-1}}\right] + M_{T_{i-1}}^2 - 2M_{T_{i-1}}\mathsf{E}\left[M_{T_i} \mid \mathcal{F}_{T_{i-1}}\right] \\
&= \mathsf{E}\left[M_{T_i}^2 \mid \mathcal{F}_{T_{i-1}}\right] + M_{T_{i-1}}^2 - 2M_{T_{i-1}}^2 \\
&= \mathsf{E}\left[M_{T_i}^2 - M_{T_{i-1}}^2 \mid \mathcal{F}_{T_{i-1}}\right] = \mathsf{E}\left[[M]_{T_i} - [M]_{T_{i-1}} \mid \mathcal{F}_{T_{i-1}}\right]
\end{aligned}$$

Hence, by this additional information, we may rewrite:

$$\begin{aligned}
\mathsf{E}\,(H \cdot M)_\infty^2 &= \sum_{i=1}^{n} \mathsf{E}\left[Z_{i-1}^2\left(M_{T_i} - M_{T_{i-1}}\right)^2\right] \\
&= \sum_{i=1}^{n} \mathsf{E}\left[Z_{i-1}^2\left([M]_{T_i} - [M]_{T_{i-1}}\right)\right] \xrightarrow{n\to\infty} \mathsf{E}\int_0^\infty H_s^2 \mathrm{d}\,[M]_s
\end{aligned}$$

For any previsible process, H, we now define the norm, $\forall$ fixed $M \in \mathcal{M}_0^2$:

$$\|H\|_M = \left(\mathsf{E}\int_0^\infty H_s^2 \,\mathrm{d}\,[M]_s\right)^{1/2}$$

Then the space:

$$L^2(M) := \{\text{previsible } H \text{ with } \|H\|_M < \infty\}$$

,(strictly speaking, this is o f course a space of certain equivalence classes) is a Hilbert space, containing $b\mathcal{E}$. Hence, stochastic integral, $(H \cdot M)$, is already defined for $H \in b\mathcal{E}$:

$$\begin{aligned} I : L^2(M) \cap b\mathcal{E} &\mapsto L^2(\mathcal{F}_\infty) \\ H &\mapsto (H \cdot M)_\infty \end{aligned}$$

, is an isometry:

$$\|H\|_M = \|I(H)\|$$

, then it extends uniquely to $L^2(M) \cap b\mathcal{E}$. We summarise into following theorem:

Theorem 3.3.2. ***(Kunita and Watanabe)***
Let $M \in \mathcal{M}_0^2$ and $H \in L^2(M)$, then the Itô integral, $(H \cdot M)_t$ is the image of H under the extension of $I : L^2(M) \mapsto L^2(\mathcal{F}_\infty)$.

Proof. Follow above discussion. □

3.4 Stopping and Localisation

Firstly, we start with our familar friends:

Definition 3.4.1. Let T be a s.t. and H be any process, define stopped process:

$$H\mathbf{1}_{(0,T]}(t) = \begin{cases} H_t, & 0 < t \le T \\ 0, & t > T \end{cases}$$

And in some of cases, we may want it to stop but not vanish, then:

Definition 3.4.2. Let T be a s.t. and X be any process, define localised process:

$$X^T(t) = \begin{cases} X(t), & 0 \le t \le T \\ X(T), & t > T \end{cases}$$

Then, if we want to extend the result to localised process, notice:

$$\begin{aligned} (H \cdot M)_t^T &= \left[Z_i\left(M_{T_i \wedge t} - M_{T_{i-1} \wedge t}\right)\right]^T = \left(Z_i \mathbf{1}_{(T_{i-1}\wedge T, T_i \wedge T]} \cdot M^T\right)_t \\ &= \left(H\mathbf{1}_{(0,T]} \cdot M^T\right)_t = Z_i\left(M_{T_i \wedge T \wedge t} - M_{T_{i-1}\wedge T \wedge t}\right) \end{aligned}$$

The next result provides the key to extending the stochastic integral by localisation:

Theorem 3.4.1. *Fix $M \in \mathcal{M}_0^2$, $\forall\, H \in L^2(M)$ and any s.t. T, we have:*

(a) $(H \cdot M)^T = \left(H\mathbf{1}_{(0,T]} \cdot M\right) = \left(H \cdot M^T\right)$,

(b) $(H \cdot M)_t^2 - \int_0^t H_s^2 \mathrm{d}\,[M]_s$ *is u.i. martingale,*

(c) If $H, K \in b\mathcal{P}$, *then* $(HK) \cdot M = H \cdot (K \cdot M) = K \cdot (H \cdot M)$,

(d) $\Delta\,(H \cdot M) = H\Delta M$,

(e) If M is continuous, then $(H \cdot M)$ is also continous.

Proof. Firstly, we prove the result for $H \in b\mathcal{E}$, and then extend it to $L^2(M)$ by the isometry. Recall that if $H^n \to H$ in $L^2(M)$, then:

$$\mathsf{E}\sup_{t\geq 0}|(H^n \cdot M)_t - (H \cdot M)_t|^2 \leq 4\mathsf{E}\,[(H^n - H) \cdot M]_\infty^2 = 4\|H^n - H\|_M^2 \to 0.$$

Hence, after passing to a subsequence if necessary, the convergence is uniform in t a.s.

(a) We first assume that, two s.t.s $U \leq V$, and $Z \in b\mathcal{F}_S$, considering:

$$H = Z\mathbf{1}_{(U,V]}$$

Then by the definition of stochastic integral for elementary previsible processes,

$$(H \cdot M)_t = Z\,(M_{t\wedge V} - M_{t\wedge U}) \implies (H \cdot M)_t^T = Z\,(M_{t\wedge V\wedge T} - M_{t\wedge U\wedge T})$$

Secondly, we could see:

$$H\mathbf{1}_{(0,T]}(t) = Z\mathbf{1}_{(U\wedge T, V\wedge T]}(t) \implies \left(H\mathbf{1}_{(0,T]} \cdot M\right) = Z\,(M_{t\wedge V\wedge T} - M_{t\wedge U\wedge T})$$

Lastly, the left one is also have same structure:

$$\left(H \cdot M^T\right)_t = Z\left(M_{V\wedge t}^T - M_{U\wedge t}^T\right) = Z\,(M_{t\wedge V\wedge T} - M_{t\wedge U\wedge T})$$

By linearity, the same conclusion holds for every $H \in b\mathcal{E}$. Hence the whole space.

(b) Firstly, notice that:

$$\mathsf{E}\sup_{t\geq 0}(H \cdot M)_t^2 \leq 4\mathsf{E}\int_0^\infty H_s^2 \mathrm{d}\,[M]_s < \infty \implies \mathsf{E}\sup_{t\geq 0}\left|(H \cdot M)_t^2 - \int_0^t H_s^2 \mathrm{d}\,[M]_s\right| < \infty$$

, then $N_t := (H \cdot M)_t^2 - \int_0^t H_s^2 \mathrm{d}\,[M]_s$ is u.i., it remains to show $\forall$ s.t. T:

$$\begin{aligned}
\mathsf{E}N_T &= \mathsf{E}\,(H \cdot M)_T^2 - \mathsf{E}\int_0^T H_s^2 \mathrm{d}\,[M]_s \\
&= \mathsf{E}\left(H\mathbf{1}_{(0,T]} \cdot M\right)_\infty^2 - \mathsf{E}\int_0^T H_s^2 \mathrm{d}\,[M]_s \\
&= \mathsf{E}\int_0^\infty \left(H\mathbf{1}_{(0,T]}(s)\right)^2 \mathrm{d}\,[M]_s - \mathsf{E}\int_0^T H_s^2 \mathrm{d}\,[M]_s = 0
\end{aligned}$$

Hence, by lemma 3.1.5, we completed (b), moreover, recall that, if $M \in \mathcal{M}_0^2$, then

$$M^2 - [M] \text{ is martingale} \implies [H \cdot M]_t = \int_0^t H_s^2 \mathrm{d}\,[M]_s\,.$$

(c) We first prove the identity for $H, K \in b\mathcal{E}$, considering $H = Z\mathbf{1}_{(S,T]}, K = Y\mathbf{1}_{(U,V]}$, then HK is again a finite linear combination of stochastic intervals, and on each such interval the increment of $K \cdot M$ is just K times the increment of M. Hence,

$$H \cdot (K \cdot M) = (HK) \cdot M.$$

By linearity, this holds for all $H, K \in b\mathcal{E}$. Similarly,

$$K \cdot (H \cdot M) = (HK) \cdot M.$$

Now let $H, K \in b\mathcal{P}$. Since H, K are bounded, all the integrals below are well-defined. Choose $H^n, K^n \in b\mathcal{E}$ such that:

$$H^n \xrightarrow{L^2} H, K^n \xrightarrow{L^2} K \implies H^n K^n \xrightarrow{L^2} HK$$

, and therefore:

$$\|H^n - H\|_{K \cdot M}^2 = \mathsf{E} \int_0^\infty (H_s^n - H_s)^2 K_s^2 \mathrm{d}\,[M]_s \to 0$$

Passing to the limit, then we completed the (c).

(d) For $H \in b\mathcal{E}$:

$$\Delta(H \cdot M)_t = \sum_{i=1}^n Z_i \mathbf{1}_{\{S_i < t \leq T_i\}} \Delta M_t = H_t \Delta M_t$$

For general $H \in L^2(M)$, take $H^n \in b\mathcal{E}$ with:

$$\|H^n - H\|_M \to 0 \implies \mathsf{E} \sum_{s \geq 0} (H_s^n - H_s)^2 (\Delta M_s)^2 \leq \mathsf{E} \int_0^\infty (H_s^n - H_s)^2 \mathrm{d}\,[M]_s \to 0$$

Thus, passing to a subsequence, the jump identity passes to the limit, and we get

$$\Delta(H \cdot M) = H \Delta M$$

(e) Follow from (d).

Therefore, all statements are proved. □

Thanks to the theorem 3.4.1's (a), we can extend:

Lemma 3.4.2. *If $H \in b\mathcal{P}$, and $M \in \mathcal{M}_{0,loc}^2$, then:*

$$(H \cdot M) \in \mathcal{M}_{0,loc}^2$$

, moreover, if $H \in b\mathcal{P}$, and $M \in c\mathcal{M}_{0,loc}^2$, then:

$$(H \cdot M) \in c\mathcal{M}_{0,loc}^2$$

Proof. Follow the result from theorem 3.4.1's (a). □

Suppose that $H \in b\mathcal{P}$,$M \in \mathcal{M}_{0,\text{loc}}^2$, and that M has paths of finite variation. We certainly want $(H \cdot M)$ to agree with the pathwise Stieltjes integral, and indeed it does.

3.5 Integration w.r.t. Local Martingales

3.6 Integration w.r.t. Continuous Semimartingales

3.7 The Itô's Formula

The Itô integral is defined by a limiting procedure far too clumsy to provide an effective means of handling such integrals in practice. Here, we shall develop the rules by which it can be manipulated, commonly called the stochastic calculus. As before, we start with stochastic calculus for bounded continuous processes, and build up via localisation.

Lemma 3.7.1. *Let $M \in c\mathcal{M}_0$ be bounded, and let V be a continuous FV_0, then:*

$$M_t^2 = \int_0^t 2M_s \mathrm{d}M_s + [M]_t$$
$$M_tV_t = \int_0^t M_s \mathrm{d}V_s + \int_0^t V_s \mathrm{d}M_s$$

Proof. The proof is straightforward, by simple algebraic manipulation:

$$\begin{aligned} M_t^2 = (M_t - M_0)^2 &= \left[\sum_{k=1}^{2^n} M_{t_k^n}^2 - M_{t_{k-1}^n}^2\right] \\ &= \sum_{k=1}^{2^n} \left(M_{t_k^n} - M_{t_{k-1}^n}\right)^2 + 2\sum_{k=1}^{2^n} \left[M_{t_{k-1}^n}\left(M_{t_k^n} - M_{t_{k-1}^n}\right)\right] \\ &\xrightarrow{n\to\infty} [M]_t + 2\int_0^t M_s \mathrm{d}M_s \end{aligned}$$

Similarly, for another one, we may see:

$$\begin{aligned} M_tV_t = M_{t_{2^n}^n}V_{t_{2^n}^n} - M_0V_0 &= \sum_{k=1}^{2^n} \left[M_{t_k^n}V_{t_k^n} - M_{t_{k-1}^n}V_{t_{k-1}^n}\right] \\ &= \sum_{k=1}^{2^n} \left[M_{t_k^n}V_{t_k^n} - M_{t_k^n}V_{t_{k-1}^n} + M_{t_k^n}V_{t_{k-1}^n} - M_{t_{k-1}^n}V_{t_{k-1}^n}\right] \\ &= \sum_{k=1}^{2^n} M_{t_k^n}\left(V_{t_k^n} - V_{t_{k-1}^n}\right) + \sum_{k=1}^{2^n} V_{t_{k-1}^n}\left(M_{t_k^n} - M_{t_{k-1}^n}\right) \\ &\xrightarrow{n\to\infty} \int_0^t M_s \mathrm{d}V_s + \int_0^t V_s \mathrm{d}M_s \end{aligned}$$

Hence, we completed the proof, and this provides us a powerful tool for I.B.P. □

The next result is the cornerstone of stochastic calculus:

Theorem 3.7.2. *(Integration by parts formula)*

Let X and Y be continuous semimartingales, then:

$$X_tY_t - X_0Y_0 = \int_0^t X_s \mathrm{d}Y_s + \int_0^t Y_s \mathrm{d}X_s + [X,Y]_t$$

Proof. Since $X, Y \in b\mathcal{P}$, then the integral is well defined, w.l.o.g., we may assume, $X_0 = Y_0 = 0$. In deed, if it is not and assume we have the above result, notice:

$$\begin{aligned} X_tY_t - X_0Y_0 &= \widetilde{X}_t\widetilde{Y}_t + X_0\widetilde{Y}_t + Y_0\widetilde{X}_t \\ &= \int_0^t \widetilde{X}_s \mathrm{d}\widetilde{Y}_s + \int_0^t \widetilde{Y}_s \mathrm{d}\widetilde{X}_s + \left[\widetilde{X}, \widetilde{Y}\right]_t + X_0\widetilde{Y}_t + Y_0\widetilde{X}_t \\ &= \int_0^t \widetilde{X}_s \mathrm{d}Y_s + \int_0^t \widetilde{Y}_s \mathrm{d}X_s + [X,Y]_t + X_0 \int_0^t \mathrm{d}Y_s + Y_0 \int_0^t \mathrm{d}X_s \\ &= \int_0^t X_s \mathrm{d}Y_s + \int_0^t Y_s \mathrm{d}X_s + [X,Y]_t \end{aligned}$$

Then assume they all start from null, if $X = M + A, Y = N + B$, and:

$$M, N \in c\mathcal{M}_{0,\mathrm{loc}}, A, B \in \mathrm{FV}_0 \implies \begin{cases} M_tN_t = \int_0^t M_s \mathrm{d}N_s + \int_0^t N_s \mathrm{d}M_s + [M,N]_t \\ A_tB_t = \int_0^t A_s \mathrm{d}B_s + \int_0^t B_s \mathrm{d}A_s \end{cases}$$

Then simple algebra gives us plus lemma 3.7.1:

$$\begin{aligned} X_tY_t &= (M_t + A_t) \times (N_t + B_t) = M_tN_t + A_tB_t + M_tB_t + N_tA_t \\ &= \int_0^t (M_s + A_s) \,\mathrm{d}\,(N_s + B_s) + \int_0^t (N_s + B_s) \,\mathrm{d}\,(M_s + A_s) + [X,Y]_t \\ &= \int_0^t X_s \mathrm{d}Y_s + \int_0^t Y_s \mathrm{d}X_s + [X,Y]_t \end{aligned}$$

Hence, we completed the proof. □

Just as in the case of finite-variation stochastic calculus, where we derived the change of variables formula (Itô's formula) from the integration-by-parts formula, so here we repeat the exercise (and the proof is essentially the same!) to derive the powerful Itô's formula for continuous semimartingales.

Theorem 3.7.3. ***(Itô's formula)***
Let $f : \mathbb{R}^d \mapsto \mathbb{R}$ be C^2, and let $X_t = \left(X_t^1, \cdots, X_t^d\right)$ be a continuous semimartingale in $\mathbb{R}^d$, (that is, each X^i is a continuous semimartingales), then:

$$f(X_t) - f(X_0) = \int_0^t \sum_i \mathsf{D}_i f(X_s) \mathrm{d}X_s^i + \frac{1}{2} \int_0^t \sum_{i,j} \mathsf{D}_{i,j} f(X_s) \mathrm{d}\left[X^i, X^j\right]_s$$

Proof. Firstly, we notice the below corollary:

Corollary 3.7.4. *Under the same condition above, $f(X)$ is continuous semimartin-*

gale, if $X = X_0 + M + A$ is the canonical decomposition of X, then:

$$f(X_t) = f(X_0) + \int_0^t \sum_i \mathsf{D}_i f(X_s) \mathrm{d}M_s^i + \left\{ \int_0^t \sum_i \mathsf{D}_i f(X_s) \mathrm{d}A_s^i + \frac{1}{2} \int_0^t \sum_{i,j} \mathsf{D}_{i,j} f(X_s) \mathrm{d}\left[M^i, M^j\right]_s \right\}$$

, then if we consider two functions: $F_t := f(X_t), G_t := g(X_t)$, by definition:

$$\begin{aligned} [F, G]_t &= \left[F^{\text{cnt mart}}, G^{\text{cnt mart}}\right]_t \\ &= \left[\int_0^{\cdot} f'(X_s) \mathrm{d}M, \int_0^{\cdot} g'(X_s) \mathrm{d}M\right] = \int_0^t f'(X_s) g'(X_s) \mathrm{d}\left[M\right]_s \end{aligned}$$

Now, using the I.B.P, (theorem 3.7.2):

$$\begin{aligned} F_t G_t - F_0 G_0 &= \int_0^t F_s \mathrm{d}G_s + \int_0^t G_s \mathrm{d}F_s + \int_0^t F_s' G_s' \mathrm{d}\left[M\right]_s \\ &= \int_0^t \left\{F_s G_s' + F_s' G_s\right\} \mathrm{d}X_s \\ &+ \frac{1}{2} \int_0^t \left\{F_s G_s'' + 2F_s' G_s' + F_s'' G_s\right\} \mathrm{d}\left[M\right]_s \end{aligned}$$

This is exactly the Itô's formula for product fg, and clearly $f(x) = x$ satisfies Itô's formula, which gives the starting points of this induction, thus all polynomials hold.

Lemma 3.7.5. *Let $f \in C^2(\mathbb{R}^d)$. For every compact set $K \subset \mathbb{R}^d$, there exists a sequence of polynomials $\{p_n\}_{n \geq 1}$ such that*

$$p_n \to f, \quad D_i p_n \to D_i f, \quad D_{ij} p_n \to D_{ij} f$$

uniformly on K, for every $1 \leq i, j \leq d$.

Inspired by above lemma, define, for any each $N > 0$:

$$U_N = \inf\left\{t \geq 0 : |X_t| + \left[M\right]_t > N\right\}$$

, then $U_N \uparrow \infty$, so it suffice to prove Itô's formula holds for $\forall\, t \leq U_N$, note that:

$$p_n(X_t) = p_n(X_0) + \int_0^t \sum_i \mathsf{D}_i p_n(X_s) \mathrm{d}X_s^i + \frac{1}{2} \int_0^t \sum_{i,j} \mathsf{D}_{i,j} p_n(X_s) \mathrm{d}\left[X^i, X^i\right]_s$$

, for $0 \leq t \leq U_N$, by letting $n \to \infty$, we will prove the limits are:

$$f(X_t) = f(X_0) + \int_0^t \sum_i \mathsf{D}_i f(X_s) \mathrm{d}X_s^i + \frac{1}{2} \int_0^t \sum_{i,j} \mathsf{D}_{i,j} f(X_s) \mathrm{d}\left[X^i, X^j\right]_s$$

(1) The first convergence is simple, since from the lemma:

$$p_n \xrightarrow{n \to \infty} f, \forall\, 0 \leq t \leq U_N$$

(2) For the continuous semimartingale term, considering its decomposition:

$$\int_0^t \sum_i \mathsf{D}_i p_n(X_s)\mathrm{d}X_s^i = \int_0^t \sum_i \mathsf{D}_i p_n(X_s)\mathrm{d}M_s^i + \int_0^t \sum_i \mathsf{D}_i p_n(X_s)\mathrm{d}A_s^i$$

(a) For FV term, it is simple since:

$$\left|\int_0^t \sum_i \mathsf{D}_i p_n(X_s)\mathrm{d}A_s^i - \int_0^t \sum_i \mathsf{D}_i f(X_s)\mathrm{d}A_s^i\right| \\ \le d \cdot \sup_i \|\mathsf{D}_i p_n - \mathsf{D}_i f\| \cdot A_t \xrightarrow{n\to\infty} 0$$

(b) For continuous martingale term, since we restrict $t \le U_N$:

$$\begin{aligned}
&\mathsf{E}\sup_{s\le t}\left|\int_0^t \sum_i \mathsf{D}_i p_n(X_s)\mathrm{d}M_s^i - \int_0^t \sum_i \mathsf{D}_i f(X_s)\mathrm{d}M_s^i\right|^2 \\
&\le 4\mathsf{E}\int_0^t \left(\sum_i \mathsf{D}_i p_n(X_s) - \sum_i \mathsf{D}_i f(X_s)\right)^2 \mathrm{d}\,[M]_s \\
&\le d \cdot \sup_{1\le i\le d} \|\mathsf{D}_i p_n - \mathsf{D}_i f\|^2 \cdot \mathsf{E}\,[M]_t \xrightarrow{n\to\infty} 0
\end{aligned}$$

, since $\mathsf{E}\,[M]_t = \mathsf{E}\,[M]_{t\wedge U_N} \le N$, taking subsequence, we have a.s. convergence.

(3) Using same way to demonstrate, by noticing, under $t \le U_N$:

$$\sum_{i,j}\left[X^i, X^j\right]_t \text{ is bounded.}$$

Hence, we completed the proof of Itô's formula. □

Remark 3.7.1. We can regard theorem 3.7.3 as a Taylor series expansion where we set:

$$\begin{aligned}
\mathrm{d}X^i\mathrm{d}X^j &:= \mathrm{d}\left[X^i, X^j\right] \\
\mathrm{d}X^i\mathrm{d}X^j\mathrm{d}X^k &:= 0
\end{aligned}$$

We shall make heavy use o f this type of differential notation. Particularly important is the fact that if X is a continuous 1-dimensional semimartingale and V is a continuous 1-dimensional finite-variation process, then:

$$\mathrm{d}X\mathrm{d}V = 0$$

Hence, if X_1, X_2, Y_1 and Y_2 are 1-dimensional continuous semimartingales such that both $X_1 - X_2$, and $Y_1 - Y_2$ are of finite variation, then:

$$\mathrm{d}X_1\mathrm{d}Y_1 = \mathrm{d}X_2\mathrm{d}Y_2$$

If X and Y are continuous semimartingales in $\mathbb{R}$, and H, K are previsible $\mathbb{R}$-valued processes, then:

$$\mathrm{d}\,(H\cdot X) = H\mathrm{d}X$$

, therefore:

$$\mathrm{d}\,[H\cdot X, K\cdot Y] = \mathrm{d}\,(H\cdot X)\,\mathrm{d}\,(K\cdot Y) = HK\mathrm{d}X\mathrm{d}Y = HK\mathrm{d}\,[X,Y]$$

Chapter 4

Beyond Itô Formula

4.1 Lévy's Theorem

Many of the most important and useful results of probability, originally proved by very complicated methods, become elegant and simple when we use Itô's formula. We begin with what is everyone's favourite proof, the Kunita-Watanabe proof of Lévy's theorem.

Theorem 4.1.1. *Let* $\{\mathsf{B}^i_t, t \geq 0\}$ *be in* $c\mathcal{M}_{0,loc}$*, for* $i = 1, 2, \cdots, n$*, if* $\forall\ i, j$*, we have:*

$$\left[\mathsf{B}^i, \mathsf{B}^j\right]_t = \begin{cases} t, & i = j \\ 0, & i \neq j \end{cases}$$

Then, $\mathsf{B}_t = (\mathsf{B}^1_t, \cdots, B^n_t)$ *is a Brownian motion on* $\mathbb{R}^n$ *with distribution* $\mathcal{N}(0, I_n)$.

Proof. Recall each distribution is captured entirely by its characteristic function, then we will show, $\forall\ s < t$, and also $\forall\ \theta \in \mathbb{R}^n$, the following holds:

$$\mathsf{E}\left[e^{i\theta(\mathsf{B}_t - \mathsf{B}_s)} \mid \mathcal{F}_s\right] = \exp\left[-\frac{1}{2}(t-s)\,|\theta|^2\right]$$

The structure is clear, then we define a function:$f : \mathbb{R}^{n+1} \mapsto \mathbb{C}$ as:

$$f(x,t) = e^{i\theta\cdot x + \frac{1}{2}|\theta|^2 t}$$

Hence, we can apply Itô formula to:

$$f(\mathsf{B}_t, t) := f(X_t)$$

, where $X_t := (\mathsf{B}^1_t, \cdots, \mathsf{B}^n_t, t)$, then we expand that into:

$$f(X_t) = f(X_0) + \int_0^t \sum_{i=1}^n \mathsf{D}_i f(X_s)\mathrm{d}\mathsf{B}^i_s + \int_0^t \mathsf{D}_{n+1} f(X_s)\mathrm{d}s$$
$$+ \frac{1}{2}\int_0^t \sum_{1\leq i,j\leq n} \mathsf{D}_{i,j} f(X_s)\mathrm{d}\left[\mathsf{B}^i, \mathsf{B}^j\right]_s + \frac{1}{2}\int_0^t \sum_{1\leq k\leq n} \mathsf{D}_{k,n+1} f(X_s)\mathrm{d}\left[\mathsf{B}^k, s\right]_s$$

, then using the information of quadratic variation:

$$\mathrm{d}\left[\mathsf{B}^k, s\right]_s = 0,\ \left[\mathsf{B}^i, \mathsf{B}^j\right]_t = \begin{cases} t, & i = j \\ 0, & i \neq j \end{cases}$$

And also by the derivative of $f(x,t)$, we reduce to:

$$
\begin{aligned}
f(\mathsf{B}_t,t)-f(\mathsf{B}_0,0) &= \int_0^t \sum_i \mathsf{D}_i f(\mathsf{B}_s,s)\mathrm{d}\mathsf{B}_s^i + \int_0^t \frac{\partial}{\partial s} f(\mathsf{B}_s,s)\mathrm{d}s \\
&+ \frac{1}{2}\int_0^t \sum_{1\le i,j\le n} \mathsf{D}_{i,j} f(X_s)\mathrm{d}\left[\mathsf{B}^i,\mathsf{B}^j\right]_s \\
&= c\mathcal{M}_{0,\text{loc}} + \int_0^t \left(e^{i\theta\mathsf{B}_s+\frac{1}{2}|\theta|^2 s}\cdot \frac{1}{2}|\theta|^2\right)\mathrm{d}s \\
&+ \int_0^t \frac{1}{2}\left[\sum_{j=1}^n e^{i\theta\mathsf{B}_s+\frac{1}{2}|\theta|^2 s}\cdot (i\theta_j)^2\right]\mathrm{d}s = c\mathcal{M}_{0,\text{loc}}
\end{aligned}
$$

Then we may conclude:

$$M_t^\theta := f(\mathsf{B}_t,t) = e^{i\theta\mathsf{B}_t+\frac{1}{2}|\theta|^2 t} = 1 + c\mathcal{M}_{0,\text{loc}} \in c\mathcal{M}_{\text{loc}}$$

Besides, we know: $\left|M_t^\theta\right| = e^{\frac{1}{2}|\theta|^2 t} < \infty$, by dominated convergence theorem:

$$\mathsf{E}\left(M_t^\theta \mid \mathcal{F}_s\right) = M_s^\theta \implies \mathsf{E}\left[e^{i\theta(\mathsf{B}_t-\mathsf{B}_s)} \mid \mathcal{F}_s\right] = e^{-\frac{1}{2}|\theta|^2(t-s)}$$

Then it tells us, $\mathsf{B}_t-\mathsf{B}_s \sim \mathcal{N}(0,(t-s)I_n)$ and independent of $\mathcal{F}_s$. $\square$

Remark 4.1.1. Lévy's theorem is the fundamental martingale characterization result, and thus it can be seen as motivation for the martingale-problem approach to diffusion theory.

4.2 c$\mathcal{M}_{0,\text{loc}}$ as time-changes of BM

A striking application o f Lévy's theorem shows that, modulo time-transformations, Brownian motion is the most general continuous local martingale. More specifically, let $M \in c\mathcal{M}_{0,\text{loc}}$, and regard $[M]_t$ as the intrinsic clock carried by M. Then M has delusions of grandeur: it thinks it is a Brownian motion!

Theorem 4.2.1. ***(Dubins and Schwarz)***
Let $M \in c\mathcal{M}_{0,loc}$, such that: $[M]_t \uparrow \infty$ as $t\to\infty$, then, $\forall\, t\ge 0$:

$$M_t = \mathsf{B}_{[M]_t}$$

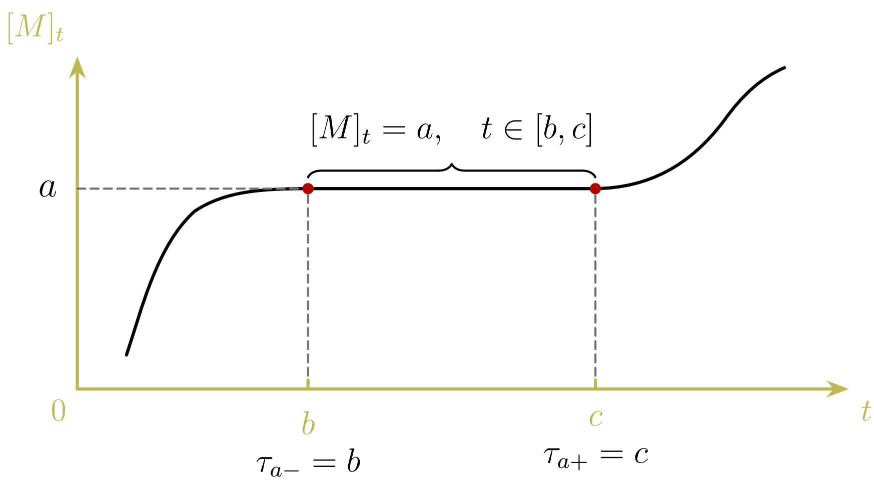
, where B is some B.M. More general, define stopping time:

$$\tau_t = \inf\left\{u \ge 0 : [M]_u > t\right\}$$

Then, $\mathsf{B}_t := M_{\tau_t}$ is a B.M w.r.t $\mathcal{G}_t := \mathcal{F}_{\tau_t}$.

Proof. (1) Firstly, for adaptivity, we know M is right-continuous, then:

$$M_{\tau_t} \in \mathcal{F}_{\tau_t} \implies \mathsf{B}_t := M_{\tau_t} \in \mathcal{F}_{\tau_t} = \mathcal{G}_t$$



(2) Next, we want to show: $t \mapsto \mathsf{B}_t$ defined above is continuous, as we can see from graph above, it suffices to show, $\forall\, q \in \mathbb{Q}^+$, define:

$$S_q = \inf \left\{ t > q : [M]_t > [M]_q \right\}$$

that $M_t = M_q$, $\forall\, t \in [q, S_q)$, note that $S_q \geq q$, then by OST:

$$\mathsf{E}\left[M_{S_q}^2 - [M]_{S_q} \mid \mathcal{F}_q \right] = M_q^2 - [M]_q$$

Since, on this interval, the constancy of $[M]$, i.e. $[M]_{S_q} = [M]_q$:

$$0 = \mathsf{E}\left[M_{S_q}^2 - M_q^2 \mid \mathcal{F}_q \right] = \mathsf{E}\left[M_{S_q}^2 - 2M_{S_q}M_q + M_q^2 \mid \mathcal{F}_q \right] = \mathsf{E}\left[\left(M_{S_q} - M_q \right)^2 \mid \mathcal{F}_q \right]$$

Then taking another expectation on both side gives us equvalence in L^2:

$$\mathsf{E}\left\{ \mathsf{E}\left[\left(M_{S_q} - M_q \right)^2 \mid \mathcal{F}_q \right] \right\} = \mathsf{E}\left(M_{S_q} - M_q \right)^2 = 0 \implies M_{S_q} = M_q$$

(3) Lastly, we want to apply Lévy's theorem, then it remains to show:

$$\mathsf{B}_t \text{ and } \mathsf{B}_t^2 - t \in c\mathcal{M}_{0,\text{loc}} \implies [\mathsf{B}]_t = t \implies \mathsf{B} \text{ is B.M.}$$

Here, we define $T_n = \inf \{ t \geq 0 : |M_t| > n \}$, and $U_n = [M]_{T_n}$, it is easy to show:

$$\tau_{t \wedge U_n} = T_n \wedge \tau_t$$

, (verify by yourself) then, we denote following:

$$\mathsf{B}_{t \wedge U_n} = \mathsf{B}_t^{U_n} = M_{\tau_{t \wedge U_n}} = M_{T_n \wedge \tau_t} = M_{\tau_t}^{T_n}$$

And by OST, we still have:

$$\mathsf{E}\left[\mathsf{B}_t^{U_n} \mid \mathcal{F}_{\tau_s} \right] = \mathsf{E}\left[M_{\tau_t}^{T_n} \mid \mathcal{F}_{\tau_s} \right] = M_{\tau_s}^{T_n} = \mathsf{B}_s^{U_n}$$

, hence in order to show $\mathsf{B} \in c\mathcal{M}_{0,\text{loc}}$, it remains to show U_n is $\{\mathcal{G}_t\}$-s.t.:

$$\{U_n \leq t\} = \left\{ [M]_{T_n} \leq t \right\} = \left\{ [M]_{T_n} \leq [M]_{\tau_t} \right\} = \{ T_n \leq \tau_t \} \in \mathcal{F}_{\tau_t} = \mathcal{G}_t$$

Finally, notice:

$$[M]_{\tau_t \wedge T_n} = [M]_{\tau_{t \wedge U_n}} = t \wedge U_n$$

, then using the fact that $M \in c\mathcal{M}_{0,\text{loc}}$:

$$\begin{aligned}\mathsf{E}\left[\left(\mathsf{B}_t^{U_n}\right)^2 - t \wedge U_n \mid \mathcal{G}_s\right] &= \mathsf{E}\left[\left(M_{\tau_t}^{T_n}\right)^2 - [M]_{\tau_t}^{T_n} \mid \mathcal{F}_{\tau_s}\right] \\ &= \left(M_{\tau_s}^{T_n}\right)^2 - [M]_{\tau_s}^{T_n} = \left(\mathsf{B}_s^{U_n}\right)^2 - s \wedge U_n\end{aligned}$$

Therefore, $(\mathsf{B}_t^2 - t)^{U_n}$ is a martingale.
For special case, we notice that:

$$M_t = M_{\tau_{[M]_t}} = \mathsf{B}_{[M]_t}$$

, hence we completed the proof. □

From the condition, we require that $[M]_\infty = \infty$, but what if not ? In this case, $[M]_\infty < \infty$:

$$\tau_t = \infty, \forall\, t \geq [M]_\infty$$

, then it is to say, $\mathsf{B}_t = M_\infty, \forall\, t \geq [M]_\infty$, which is clearly not a B.M, so how to fix it ? The natural idea is that, we may piece togather another B.M. at the end, such that:

$$\mathsf{B}_t = \begin{cases} M_{\tau_t}, & t < [M]_\infty \\ M_\infty + W_{(t-[M]_\infty)}, & t \geq [M]_\infty \end{cases}$$

, where W is a s.B.M. independent of M.But whether the limit exists ?

Lemma 4.2.2. *On $\{[M]_\infty < \infty\}$, the limit:*

$$M_\infty := \lim_{t \uparrow \infty} M_t \;\exists\; a.s.$$

Proof. Define:

$$S_k = \inf\{t \geq 0 : [M]_t > k\}$$

, then on $\{[M]_\infty < \infty\}$,

$$S_k = \infty, \forall\, k > [M]_\infty$$

Since, M^{S_k} is a L^2-bounded martingale, then:

$$\lim_{t \uparrow \infty} M_t^{S_k} \;\exists\; \text{a.s.}$$

Now, we send $k \uparrow \infty$, to get our result. □

Lastly, by enlargement of the probability space $(\Omega, \mathcal{F}, \mathsf{P})$:

$$\tilde{\Omega} = \Omega \otimes \Omega^*$$
$$\tilde{\mathcal{F}} = \mathcal{F} \otimes \mathcal{F}^*$$
$$\tilde{\mathsf{P}} = \mathsf{P} \otimes \mathsf{P}^*$$

, where on $(\Omega^*, \mathcal{F}^*, \mathcal{F}_t^*, \mathsf{P}^*)$, we have W is B.M. independent of M, and on this enriched probability space, we fixed with the general case, to conclude:

Theorem 4.2.3. *Let $M \in c\mathcal{M}_{0,loc}$, then $\exists$ B.M., B, s.t.:*

$$M_t = \mathsf{B}_{[M]_t}, \forall\, t \geq 0$$

Proof. Follow above discussion. □

4.3 Exponential semimartingales

The next application of Itô's formula is of central importance because it yields a rare example of a stochastic differential equation with an explicit solution, we can discuss it outside the framework of the general theory of stochastic differential equations.

Theorem 4.3.1. ***(Doléans's Theorem)***
Let X be a continuous semimartingale with $X_0 = 0$. Suppose $Z_0 \in \mathcal{F}_0$, then there exists a unique continuous semimartingale Z s.t.:

$$Z_t = Z_0 + \int_0^t Z_s \mathrm{d}X_s$$

, the solution is:

$$Z_t = Z_0 \exp\left(X_t - \frac{1}{2}[X]_t\right)$$

Remark 4.3.1. Before we move into the proof, we need to clearify that even though we have not touched SDE theory, but we could be motivated to see why it has such form:

$$\mathrm{d}Z_t = Z_t\mathrm{d}X_s \approx \mathrm{d}y = y\mathrm{d}x \implies y = ce^x$$

, and in order to proceed, we may make following notataion:

$$\mathcal{E}(X)_t := \exp\left(X_t - \frac{1}{2}[X]_t\right)$$

, is called the <u>**expoential semimartingale**</u> of X.

Proof. Considering its expoential semimartingale as:

$$\mathcal{E}(X)_t = \exp\left(X_t - \frac{1}{2}[X]_t\right) = f(X_t, [X]_t)$$

Then we can apply Itô's formula to $f(x,y) = e^{x-\frac{1}{2}y}$:

$$\begin{aligned}\mathrm{d}\mathcal{E}(X)_t &= \mathrm{d}f(X_t,[X]_t) = \partial_X f(X_t,[X]_t)\mathrm{d}X_t + \partial_{[X]} f(X_t,[X]_t)\mathrm{d}[X]_t + \frac{1}{2}\partial_{XX} f(X_t,[X]_t)\mathrm{d}[X]_t \\ &= \mathcal{E}(X)_t\mathrm{d}X_t + \left(-\frac{1}{2}\right)\mathcal{E}(X)_t\mathrm{d}[X]_t + \frac{1}{2}\mathcal{E}(X)_t\mathrm{d}[X]_t = \mathcal{E}(X)_t\mathrm{d}X_t\end{aligned}$$

Hence, $\mathcal{E}(X)_t$ is a solution to:

$$\mathrm{d}Z_t = Z_t\mathrm{d}X_t$$

Next is to show the uniqueness, assume Z is another solution, we will show:

$$Z_t = Z_0 \mathcal{E}(X)_t$$

Since we have have: $\mathrm{d}Z_t = Z_t \mathrm{d}X_t$, then we define:

$$Y_t = \exp\left(-X_t + \frac{1}{2}[X]_t\right) = \left(\mathcal{E}(X)_t\right)^{-1}$$

It suffices to prove,

$$\mathrm{d}(Z_t Y_t) = 0 \implies Z_t Y_t = Z_0 Y_0 = Z_0$$

, which is simple to show by noticing:

$$\mathrm{d}Y_t = (-1) Y_t \mathrm{d}X_t + \frac{1}{2} Y_t \mathrm{d}[X]_t + \frac{1}{2}(-1)^2 Y_t \mathrm{d}[X]_t = -Y_t \mathrm{d}X_t + Y_t \mathrm{d}[X]_t$$

Lastly, since I.B.P:

$$\begin{aligned}
\mathrm{d}(Z_t Y_t) &= Z_t \mathrm{d}Y_t + Y_t \mathrm{d}Z_t + \mathrm{d}[Y_t, Z_t] \\
&= Z_t\left(-Y_t \mathrm{d}X_t + Y_t \mathrm{d}[X]_t\right) + Y_t\left(Z_t \mathrm{d}X_t\right) + \left(-Y_t Z_t\right)\mathrm{d}[X]_t = 0
\end{aligned}$$

Then we completed the proof. □

Remark 4.3.2. The last step, we used the result from homework:

$$\mathrm{d}[H \cdot X, K \cdot Y]_t = HK \mathrm{d}[X, Y]_t$$

In general, let $M \in c\mathcal{M}_{0,\mathrm{loc}}$, then:

$$\mathcal{E}(\theta M)_t = \exp\left(\theta M_t - \frac{1}{2}\theta^2 [M]_t\right) \in c\mathcal{M}_{\mathrm{loc}}$$

Moreover, by Fatou's Lemma, we can see it is also a supermartingale:

$$\begin{aligned}
\mathsf{E}\left[\mathcal{E}(\theta M)_t \mid \mathcal{F}_s\right] &= \mathsf{E}\left[\lim_{n\to\infty} \mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s\right] \\
&\leq \liminf_{n\to\infty} \mathsf{E}\left[\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s\right] = \liminf_{n\to\infty} \mathcal{E}(\theta M)_s^{T_n} = \mathcal{E}(\theta M)_s
\end{aligned}$$

, with bounded expectation: $\mathsf{E}\mathcal{E}(M)_t \leq 1$. For more information, we want to estimate it:

Theorem 4.3.2. *If for some $t > 0$, $\exists\, K_t > 0$, s.t.:*

$$[M]_t < K_t \ a.s.$$

, then for this $t > 0$, and $\forall\, y \geq 0$:

$$\mathsf{P}\left(\max_{s \leq t} M_s > y\right) \leq \exp\left(-\frac{y^2}{2K_t}\right)$$

Proof. By using the fact $\mathcal{E}(\theta M)$ is supermartingale, and Doob's inequality:

$$\begin{aligned}\mathsf{P}\left(\max_{s\le t} M_s > y\right) &= \mathsf{P}\left(\max_{s\le t}\mathcal{E}(\theta M)_s > \exp\left(\theta y - \frac{1}{2}\theta^2 K_t\right)\right)\\ &\le \frac{\mathsf{E}\mathcal{E}(\theta M)_t}{\exp\left(\theta y - \frac{1}{2}\theta^2 K_t\right)} \le \exp\left(-\theta y + \frac{1}{2}\theta^2 K_t\right) \le \exp\left(-\frac{y^2}{2K_t}\right)\end{aligned}$$

The last step is to pick θ so that we minimise the exponent. □

And if we restrict the condition, we may gain more powerful result:

Theorem 4.3.3. *Suppose* $\forall\, t > 0, \exists\, K_t > 0$, *s.t.:*

$$[M]_t < K_t \ a.s.$$

, then $\mathcal{E}(\theta M)$ *is a true martingale.*

Proof. Firstly, we define:

$$Z_t = \max_{s\le t} M_s$$

, and by theorem 4.3.2, notice that, $\forall\, \theta > 0$:

$$\begin{aligned}\mathsf{E}e^{\theta Z_t} &= 1 + \mathsf{E}\int_0^{Z_t}\theta e^{\theta y}\mathrm{d}y = 1 + \mathsf{E}\int_0^{\infty}\mathbf{1}_{\{Z_t>y\}}\theta e^{\theta y}\mathrm{d}y\\ &= 1 + \int_0^{\infty}\theta e^{\theta y}\mathsf{P}(Z_t > y)\,\mathrm{d}y \le 1 + \int_0^{\infty}\theta e^{\theta y}e^{-\frac{y^2}{2K_t}}\mathrm{d}y < \infty\end{aligned}$$

Then using the fact $\mathcal{E}(\theta M)_t \in c\mathcal{M}_{\mathrm{loc}}$, we get $\exists\, T_n \uparrow \infty$, s.t., $\forall\, s < t$:

$$\mathsf{E}\left[\mathcal{E}(\theta M)_t^{T_n} \mid \mathcal{F}_s\right] = \mathcal{E}(\theta M)_s^{T_n}$$

Lastly, since $\mathcal{E}(\theta M)_t^{T_n} \le e^{\theta Z_t}$ and $\mathsf{E}e^{\theta Z_t} < \infty$, using B.C.T we completed the proof. □

As natural extension, we can check the special case on B.M.

Corollary 4.3.4. *Let* B *be B.M. in* $\mathbb{R}^n$*, let* H *be bounded previsible process, then:*

$$\exp\left(\int_0^t H\mathrm{d}\mathsf{B} - \frac{1}{2}\int_0^t |H_s|^2\,\mathrm{d}s\right)$$

, is a true martingale.

Proof. Directly follow from theorem 4.3.3. □

Remark 4.3.3. If we take, $M = \int_0^t H\mathrm{d}\mathsf{B}$, then:

$$[M]_t = \int_0^t |H_s|^2\,\mathrm{d}s < K_t$$

Corollary 4.3.5. *Let $M \in c\mathcal{M}_{0,loc}$, then $\forall\, t > 0, y > 0, K > 0$:*

$$\mathsf{P}\left(\max_{s\le t} M_s \ge y, [M]_t \le K\right) \le \exp\left(-\frac{y^2}{2K}\right)$$

This also implies:

$$\mathsf{P}\left(\max_{s\le t} |M_s| \ge y, [M]_t \le K\right) \le 2\exp\left(-\frac{y^2}{2K}\right)$$

Proof. Let $T = \inf\{u : [M]_u \ge K\}$, then:

$$\mathsf{P}\left(\max_{s\le t} M_s \ge y, [M]_t \le K\right) \le \mathsf{P}\left(\max_{s\le t} M_s^T \ge y\right) \le \exp\left(-\frac{y^2}{2K}\right)$$

The last step it apply theorem 4.3.2 to M^T. □

Lastly, we enclose this section by another theorem without proof:

Theorem 4.3.6. ***(Novikov's Theorem)***
Let $M \in c\mathcal{M}_{loc}$, if $\forall\, t > 0$, $\mathsf{E}e^{\frac{1}{2}[M]_t} < \infty$, then $\mathcal{E}(M)_t$ is true martingale.

Proof. See p.198 of KS. □

4.4 Burkholder-Davis-Gundy Inequalities

Martingale inequalities of diverse and wondrous forms have been the subject of intensive investigation, and it would be impossible to attempt a detailed coverage. But B-D-G inequality worth its own place, and firstly, we need to introduce one concept:

Definition 4.4.1. We say a function $F : \mathbb{R}^+ \mapsto \mathbb{R}^+$ is **moderate** if:

(a) it is continuous,

(b) it is increasing,

(c) $F(x) = 0 \Leftrightarrow x = 0$,

(d) $\forall\, \alpha > 1$:

$$\sup_{x>0} \frac{F(\alpha x)}{F(x)} < \infty$$

The most obvious examples of moderate F are the powers:

$$F(x) = x^p, \forall\, p > 0$$

, and in some of cases, we may want to use logarithm to control, then:

$$F(x) = x^p \log(x \vee 1), \forall\, p > 0$$

Now, we can introduce the B-D-G inequality:

Theorem 4.4.1. ***(Burkholder-Davis-Gundy Inequalities)***
Let F be moderate, then $\exists$ universal constants, $c_F \leq C_F < \infty$, s.t., $\forall\, M \in c\mathcal{M}_{0,loc}$:

$$c_F \mathsf{E} F\left([M]_\infty^{\frac{1}{2}}\right) \leq \mathsf{E} F\left(M_\infty^*\right) \leq C_F \mathsf{E} F\left([M]_\infty^{\frac{1}{2}}\right)$$

, where $M_t^* = \sup_{s \leq t} |M_s|$. Generally, fix $t > 0$, let $\widetilde{M}_s^t = M_{s \wedge t}$, then:

$$c_F \mathsf{E} F\left([M]_t^{\frac{1}{2}}\right) \leq \mathsf{E} F\left(M_t^*\right) \leq C_F \mathsf{E} F\left([M]_t^{\frac{1}{2}}\right)$$

Remark 4.4.1. For the case $F(x) = x^p$, Getoor and Sharpe give a nice proof using stochastic calculus. We follow Burkholder's proof which relies on the good λ inequality.

Then we postpone the proof of B-D-G inequalities, and introduce an auxiliary lemma:

Lemma 4.4.2. ***("good" λ inquality)***
Let X and Y be nonnegative r.v.s, suppose $\exists\, \beta > 1$, s.t. $\forall\, \lambda > 0, \delta > 0$:

$$\mathsf{P}\left(X > \beta\lambda, Y \leq \delta\lambda\right) \leq \psi\left(\delta\right) \mathsf{P}\left(X > \lambda\right)$$

, where $\psi(\delta) \downarrow 0$ as $\delta \downarrow 0$. Then for each moderate F, $\exists\, C = C(\beta, \psi, F) > 0$, s.t.:

$$\mathsf{E} F\left(X\right) \leq C \mathsf{E} F\left(Y\right)$$

Proof. W.l.o.g., we may assume $\mathsf{E} F(Y) < \infty$, similarly we assume the good λ inequality holds, then it also holds true, if we replace X with $X \wedge a$, $\forall\, a > 0$:

(1) If $a \leq \beta\lambda$, then:

$$X \wedge a \leq a \leq \beta\lambda \implies \mathsf{P}\left(X > \beta\lambda, Y \leq \delta\lambda\right) = 0$$

(2) If $a > \beta\lambda > \lambda$, then:

$$\begin{matrix} X \wedge a > \beta\lambda \Leftrightarrow X > \beta\lambda \\ X \wedge a > \lambda \Leftrightarrow X > \lambda \end{matrix} \implies \begin{matrix} \mathsf{P}\left(X \wedge a > \beta\lambda, Y \leq \delta\lambda\right) = \mathsf{P}\left(X > \beta\lambda, Y \leq \delta\lambda\right) \\ \mathsf{P}\left(X \wedge a > \lambda\right) = \mathsf{P}\left(X > \lambda\right) \end{matrix}$$

Hence, we may also assume $\mathsf{E} F(X) < \infty$, since if not, we send $a \uparrow \infty$, and:

$$\mathsf{E} F\left(X\right) = \lim_{a \uparrow \infty} \mathsf{E} F\left(X \wedge a\right)$$

Now, notice the moderate property, then:

$$\sup_{x > 0} \frac{F(x)}{F(x/\beta)} < \infty \implies \exists\, \gamma > 0, \text{ s.t. } F(x) \leq \frac{1}{\gamma} F(x/\beta), \forall\, x > 0$$

Now, we can integrate on both side:

$$\int F(\mathrm{d}\lambda)\, \psi(\delta)\, \mathsf{P}\left(X > \lambda\right) \geq \int F(\mathrm{d}\lambda)\, \mathsf{P}\left(X > \beta\lambda, Y \leq \delta\lambda\right)$$

Now, we may check one by one:

(a) For LHS:

$$\begin{aligned}\int F(\mathrm{d}\lambda)\,\psi(\delta)\,\mathsf{P}(X>\lambda) &= \psi(\delta)\int F(\mathrm{d}\lambda)\,\mathsf{P}(X>\lambda)\\ &= \psi(\delta)\int F(\mathrm{d}\lambda)\,\mathsf{E}\mathbf{1}_{\{X>\lambda\}}\\ &= \psi(\delta)\,\mathsf{E}\int F(\mathrm{d}\lambda)\,\mathbf{1}_{\{X>\lambda\}}\\ &= \psi(\delta)\,\mathsf{E}\int_0^X F(\mathrm{d}\lambda) = \psi(\delta)\,\mathsf{E}F(X)\end{aligned}$$

(b) For RHS:

$$\begin{aligned}\int F(\mathrm{d}\lambda)\,\mathsf{P}(X>\beta\lambda, Y\le\delta\lambda) &= \int F(\mathrm{d}\lambda)\,\mathsf{E}\mathbf{1}_{\left\{\frac{Y}{\delta}\le\lambda\le\frac{X}{\beta}\right\}}\\ &= \mathsf{E}\left[F(X/\beta)-F(Y/\delta)\right] \ge \mathsf{E}\gamma F(X)-\mathsf{E}F(Y/\delta)\end{aligned}$$

Now, since $\psi(\delta)$ can be arbitrarly small, then let $\delta>0$ be small s.t.:

$$\psi(\delta)<\frac{\gamma}{2} \implies \gamma-\psi(\delta)\ge\frac{\gamma}{2}>0$$

Lastly, we use the same trick to find a parameter for Y, i.e. using:

$$\sup_{y>0}\frac{F(y/\delta)}{F(y)}<\infty \implies \exists\,\mu>0,\ \text{s.t. } \mu F(y)\ge F(y/\delta),\forall\, y>0$$

Hence, we can combine the two side togather:

$$\mu\mathsf{E}F(Y)\ge \mathsf{E}F(Y/\delta)\ge(\gamma-\psi(\delta))\,\mathsf{E}F(X)\ge\frac{\gamma}{2}\mathsf{E}F(X)$$

Then, we completed the proof, by solving out $C:=2\mu/\gamma$. □

Using lemma 4.4.2, we can proceed:

Proof. of theorem 4.4.1.
The key idea is to use lemma 4.4.2, define $\tau=\inf\{u\ge 0:|M_u|>\lambda\}$, notice that:

$$\begin{aligned}\mathsf{P}\left(M_\infty^*>\beta\lambda,[M]_\infty^{\frac12}\le\delta\lambda\right) &= \mathsf{P}\left(M_\infty^*>\beta\lambda,[M]_\infty^{\frac12}\le\delta\lambda,\tau<\infty\right)\\ &= \mathsf{E}\left[\mathbf{1}_{\{\tau<\infty\}}\mathsf{P}\left(M_\infty^*>\beta\lambda,[M]_\infty^{\frac12}\le\delta\lambda\mid\mathcal{F}_\tau\right)\right]\end{aligned}$$

To better use the gambler's ruin, we define:

$$N_t=(M_{t+\tau}-M_\tau)^2-\left([M]_{t+\tau}-[M]_\tau\right)\in c\mathcal{M}_{0,\mathrm{loc}}$$

Besides, on $\left\{M_\infty^*>\beta\lambda,[M]_\infty^{\frac12}\le\delta\lambda\right\}$, we have $\exists\,\tilde t>0$, s.t.:

$$M_{\tilde t+\tau}\ge\beta\lambda \implies N_{\tilde t}=(M_{\tilde t+\tau}\pm\lambda)^2-\left([M]_{\tilde t+\tau}-[M]_{\tilde t}\right)\ge\left[(\beta-1)\lambda\right]^2-(\delta\lambda)^2$$

Moreover, since $N_t \geq -\delta^2\lambda^2, \forall\, t \geq 0$, hence N_t reaches $[(\beta-1)\,\lambda]^2 - (\delta\lambda)^2$ before $-\delta^2\lambda^2$:

$$\begin{aligned}\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \leq \delta\lambda \mid \mathcal{F}_\tau\right) &\leq \mathsf{P}\left(\tau_{[(\beta-1)\lambda]^2-(\delta\lambda)^2} - \tau_{-\delta^2\lambda^2}\right) \\ &\leq \frac{(\delta\lambda)^2}{[(\beta-1)\,\lambda]^2} = \frac{\delta^2}{(\beta-1)^2}\end{aligned}$$

, here we take, $0 < \delta < \beta - 1$, therefore:

$$\mathsf{P}\left(M_\infty^* > \beta\lambda, [M]_\infty^{\frac{1}{2}} \leq \delta\lambda\right) \leq \frac{\delta^2}{(\beta-1)^2}\mathsf{P}\,(\tau < \infty) = \frac{\delta^2}{(\beta-1)^2}\mathsf{P}\,(M_\infty^* > \lambda)$$

For another part, we similarly define:

$$\widetilde{N}_t = [M]_{t+\tilde{\tau}} - [M]_{\tilde{\tau}} - (M_{t+\tilde{\tau}} - M_{\tilde{\tau}})^2$$

, with: $\tilde{\tau} = \int \left\{u \geq 0 : [M]_u^{\frac{1}{2}} > \lambda\right\}$. Then by lemma 4.4.2 we completed the proof. □

4.5 Semimartingale local time

The applications of Itô's formula we have seen so far have been valid for continuous semimartingales with values in $\mathbb{R}^n$ for any $n \geq 1$. For 1-dimensional continuous semimartingales, we have the added great advantage of having the concept of local time. Itô's formula and local time together supply simple and beautiful existence, uniqueness, and comparison results for 1-dimensional SDEs, and the combination of local time and Itô excursion theory can be applied very eflectively to 1-dimensional diffusions.
For a B.M. $\{\mathsf{B}_t, t \geq 0\}$, we consider its local time at 0 as:

$$\ell_t^0 = \int_0^t \delta_0\,(\mathsf{B}_s)\,\mathrm{d}s$$

, where δ_0 is the Dirac delta function, s.t., $\forall$ function f:

$$\int f(x)\;\delta_0\,(x)\,\mathrm{d}x = f(0)$$

, which is actually a distribution, here we may imagine a r.v. ξ, s.t. $\mathsf{P}\,(\xi = 0) = 1$, then:

$$\mathsf{E}f(\xi) = \int f(x)\;\delta_0\,(x)\,\mathrm{d}x = f(0)$$

Since, clearly, this is increasing, then it can induce a meausre as:

$$\mu\,([a,b)) = \ell_b^0 - \ell_a^0$$

, and its support is, for sure, that:

$$\mathrm{Supp}\,(\mu) = \{t \geq 0 : \mathsf{B}_t = 0\}$$

And generally, we call the **level set**: $\{t \geq 0 : \mathsf{B}_t = x\}$, and its local time:

$$\ell_t^x = \int_0^t \delta_x\,(\mathsf{B}_s)\,\mathrm{d}s$$

To be rigourous about this concept, we propose the Tanaka's formula:

Theorem 4.5.1. ***(Tanaka's formula)***
Let X be a continuous semimartingale, then $\exists$ continuous increasing adapted process, $\{\ell_t, t \geq 0\}$, such that:

$$|X_t| - |X_0| = \int_0^t sgn\,(X_s)\,\mathrm{d}X_s + \ell_t$$

, where $sgn\,(x) = \begin{cases} 1, & x > 0 \\ -1, & x \leq 0 \end{cases}$, then ℓ is called the ***local time*** *of X at 0, and:*

$$\int_0^t \mathbf{1}_{\{X_s \neq 0\}} \mathrm{d}\ell_s = 0$$

Remark 4.5.1. We postpone the poof, and check some interesting points:

(a) Intuitively, consider the function, $f(x) = |x|$, then basic calculus gives us:

$$f'(x) = \mathrm{sgn}\,(x)$$

, at $x \neq 0$, and if we are able to do another derivative, then:

$$f''(x) \stackrel{?}{=} 2\delta_0\,(x)$$

since this is not legal in general, then we call it, **distributional derivative**:

Definition 4.5.1. We say $\frac{\mathrm{d}}{\mathrm{d}x}T(t)$ is the **distributional derivative** of $T(x)$, if $\forall$ smooth function, $\phi(x)$, with $\lim_{|x|\to\infty} \phi(x) = 0$, then:

$$\int \left(\frac{\mathrm{d}}{\mathrm{d}x}T(x)\right)\phi(x)\mathrm{d}x = -\int \phi'(x)T(x)\mathrm{d}x$$

It is easy to see, it is the same as I.B.P, with the test function, ϕ, be null at the boundary. Then we also can see why the distribution derivative of $\mathrm{sgn}(x)$:

$$\begin{aligned} 2\phi(0) &= \int_{\mathbb{R}} 2\delta_0\,(x)\,\phi(x)\mathrm{d}x = -\int_{\mathbb{R}} \phi'(x)\,\mathrm{sgn}\,(x)\,\mathrm{d}x \\ &= -\int_0^\infty \phi'(x)\mathrm{d}x + \int_{-\infty}^0 \phi'(x)\mathrm{d}x = 2\phi(0) \end{aligned}$$

(b) If we do a simple transformation, we recover the "level set" process:

$$|X_t - x| - |X_0 - x| = \int_0^t \mathrm{sgn}\,(X_s - x)\,\mathrm{d}X_s + \ell_t^x$$

(c) Another important observation is that:

$$\begin{aligned} |X_t| &= X_t^+ + X_t^- \\ X_t &= X_t^+ - X_t^- \end{aligned} \implies 2X_t^+ = |X_t| + X_t$$

Then, we may use theorem 4.5.1 to expand $|X_t|$:

$$2X_t^+ = |X_t| + X_t = \underbrace{|X_0| + \int_0^t \mathbf{1}_{\{X_s>0\}} \mathrm{d}X_s - \int_0^t \mathbf{1}_{\{X_s\leq 0\}} \mathrm{d}X_s + \ell_t}_{|X_t|} + \underbrace{\int_0^t \mathbf{1} \cdot \mathrm{d}X_s + X_0}_{X_t}$$

Then if we rearrange the terms, we gain the expansion of X_t^+:

$$X_t^+ - X_0^+ = \int_0^t \mathbf{1}_{\{X_s>0\}} \mathrm{d}X_s + \frac{1}{2}\ell_t$$

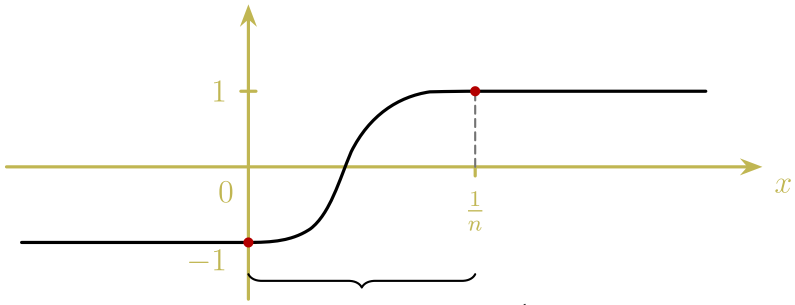
Similarly, by $2X_t^- = |X_t| - X_t$, we also have:

$$X_t^- - X_0^- = -\int_0^t \mathbf{1}_{\{X_s\leq 0\}} \mathrm{d}X_s + \frac{1}{2}\ell_t$$

Proof. Motivated by the observation that, if we can apply Itô's formula to $f(x) = |x|$:

$$\begin{aligned} |X_t| &= |X_0| + \int_0^t f'(X_s)\mathrm{d}X_s + \frac{1}{2}\int_0^t f''(X_s)\mathrm{d}\,[X]_s \\ &= |X_0| + \int_0^t \operatorname{sgn}(X_s)\,\mathrm{d}X_s + \int_0^t \delta_0(X_s)\,\mathrm{d}\,[X]_s \end{aligned}$$

And for B.M. $\mathrm{d}\,[X]_s$ becomes $\mathrm{d}s$, and above gives more general definition of local time, but the issue is we cannot do so, as the absolute function is not second differentiable. Then our idea is to find a series of function, whose properties is good enough, so that we can interchange the limit and integral. Starting from $\operatorname{sgn}(\cdot)$, considering:



smooth transition on $(0, \frac{1}{n})$

$$\rho(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases} \implies \varphi_n(x) = \begin{cases} -1, & x \leq 0 \\ -1 + 2\dfrac{\rho(x)}{\rho(x) + \rho(1/n - x)}, & 0 < x < 1/n \\ 1, & x \geq 1/n \end{cases}$$

Clearly, $\varphi_n \in C^\infty(\mathbb{R})$ and $\operatorname{sgn}(x) = \lim_{n\to\infty} \varphi_n(x)$, and then we could define:

$$f_n(x) = \begin{cases} -x, & x \leq 0 \\ \int_0^x \varphi_n(t)\mathrm{d}t, & x > 0 \end{cases} \implies f_n'(x) = \varphi_n(x), |x| = \lim_{n\to\infty} f_n(x)$$

Besides, it is also uniformly bounded, since:

$$|f(x) - f_n(x)| = \left|\int_0^{|x|} [1 - \varphi_n(t)]\,\mathrm{d}t\right| \leq \int_0^{|x|} |1 - \varphi_n(t)|\,\mathrm{d}t \leq \int_0^{1/n} |1 - \varphi_n(t)|\,\mathrm{d}t \leq 2 \cdot \frac{1}{n}$$

Now we shall apply the Itô formula to f_n to gain:

$$f_n(X_t) = f_n(X_0) + \int_0^t f_n'(X_s)\mathrm{d}X_s + \frac{1}{2}\int_0^t f_n''(X_s)\mathrm{d}\left[X\right]_s$$

Here, we know from the construction that:

$$f(X_0) = \lim_{n\to\infty} f_n(X_0), \text{ and } f(X_t) = \lim_{n\to\infty} f_n(X_t)$$

Then it suffice to show the convergence of first derivative term, since if it holds:

$$\frac{1}{2}\int_0^t f_n''(X_s)\mathrm{d}\left[X\right]_s =: C_t^n \xrightarrow{\text{a.s.}} \ell_t := |X_t| - |X_0| - \int_0^t \mathrm{sgn}\left(X_s\right)\mathrm{d}X_s$$

, as the other terms already converged, and for first derivative term, notice:

$$\int_0^t f_n'(X_s)\mathrm{d}X_s = \int_0^t f_n'(X_s)\mathrm{d}M_s + \int_0^t f_n'(X_s)\mathrm{d}A_s$$

Then we shall consider them one by one:

(1) For the FV term, notice that

$$\begin{aligned}\left|\int_0^t \left[f'(X_s) - f_n'(X_s)\right]\mathrm{d}A_s\right| &= \left|\int_0^t \left[\mathrm{sgn}\left(X_s\right) - \varphi_n\left(X_s\right)\right]\mathrm{d}A_s\right| \\ &\leq \int_0^t \left|\mathrm{sgn}\left(X_s\right) - \varphi_n\left(X_s\right)\right| \cdot \left|\mathrm{d}A_s\right|\end{aligned}$$

Then since, $\varphi_n(x) \uparrow \mathrm{sgn}(x)$, $\forall\, x$, by D.C.T:

$$\int_0^t \varphi_n\left(X_s\right)\mathrm{d}A_s \xrightarrow{\text{a.s.}} \int_0^t \mathrm{sgn}\left(X_s\right)\mathrm{d}A_s$$

, uniformly in t.

(2) For martingale term, we need to bound it first, define:

$$T_k = \inf\left\{t \geq 0 : |M_t| \geq k\right\} \implies M^{T_k} \text{ is bounded}$$

Then we may assume M is bounded, hence:

$$\left\|\int_0^\infty \left[\mathrm{sgn}\left(X_s\right) - \varphi_n\left(X_s\right)\right]\mathrm{d}M_s\right\|_2^2 = \mathsf{E}\int_0^\infty \left[\mathrm{sgn}\left(X_s\right) - \varphi_n\left(X_s\right)\right]^2 \mathrm{d}\left[M\right]_s \longrightarrow 0$$

Then by Doob's inequality, we also have:

$$\mathsf{E}\sup_{t\leq T}\left|\int_0^\infty \left[\mathrm{sgn}\left(X_s\right) - \varphi_n\left(X_s\right)\right]\mathrm{d}M_s\right|^2 \longrightarrow 0$$

Hence, from L^2 convergence, we may have a subsequence so that:

$$\sup_{t\geq 0}\int_0^t \left[\mathrm{sgn}\left(X_s\right) - \varphi_n\left(X_s\right)\right]\mathrm{d}M_s \xrightarrow{\text{a.s.}} 0$$

Lastly, we remain to show the support, since we see C_t^n as distribution function, then

$$\mathrm{d}C_s^n = \frac{1}{2} f_n''(X_s) \mathrm{d}[X]_s$$

Therefore, $\forall \gamma > 0$, if $1/n < \gamma$, then:

$$|X_s| > \gamma > \frac{1}{n} \implies f_n''(X_s) = 0 \implies \int_0^t \mathbf{1}_{\{|X_s|>\gamma\}} \mathrm{d}C_s^n = 0$$

Hence, we splite into smaller interval and sen it to infinity, i.e.:

$$\int_0^t \mathbf{1}_{\{|X_s|>\gamma\}} \mathrm{d}\ell_s \leq \lim_{n\to\infty} \int_0^t \mathbf{1}_{\{|X_s|>\gamma/2\}} \mathrm{d}C_s^n = 0$$

Then this completes the proof since γ can be arbitrary. □

4.6 Stratonovich Calculus

All above we are discussing is Itô calculus, it is powerful, and there are lots of results derived from it, the last section aims to remind reader that there are not only one way to define the stochastic integral, and the key is how the point is picked.

Definition 4.6.1. Let X and Y be continuous semimartingale, then the **Stratonovich integral** is defined by:

$$\int_0^t Y_s \partial X_s := \int_0^t Y_s \mathrm{d}X_s + \frac{1}{2}[X, Y]_t$$

Remark 4.6.1. Throught out this section, we always use:

d to denote the Itô differential
∂ to denote the Stratonovich differential

The intuition of why we define like this is that, recall:

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \int_0^t X_s \mathrm{d}Y_s + \int_0^t Y_s \mathrm{d}X_s + [X,Y]_t \\ &= \underbrace{\int_0^t X_s \mathrm{d}Y_s + \frac{1}{2}[X,Y]_t}_{\int_0^t X_s \partial Y_s} + \underbrace{\int_0^t Y_s \mathrm{d}X_s + \frac{1}{2}[X,Y]_t}_{\int_0^t Y_s \partial X_s} \end{aligned}$$

Then since the symmetry of cross variation, we may give each term one half cross variation to make the intergral "clean", benefiting from that, the differential form is also clean:

Theorem 4.6.1. ***(Stratonovich formula)***
Let $f \in C^3(\mathbb{R})$, then:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) \partial X_s$$

Proof. By Itô formula:

$$f(X_t) = f(X_0) + \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\int_0^t f''(X_s)\,\mathrm{d}\left[X\right]_s$$

From the definition of Stratonovich integral:

$$\int_0^t f'(X_s)\partial X_s = \int_0^t f'(X_s)\,\mathrm{d}X_s + \frac{1}{2}\left[f'(X), X\right]_t$$

And to see the cross variation of f', we apply Itô formula to f':

$$\mathrm{d}f'(X_t) = f''(X_t)\mathrm{d}X_t + \frac{1}{2}f'''(X_t)\mathrm{d}\left[X\right]_t \implies \left[f'(X), X\right]_t = \int_0^t f''(X_s)\,\mathrm{d}\left[X\right]_s$$

Therefore, we recovered the term and also completed the proof. $\square$

From another point of view, the Riemann-Sum approximation, we could see the difference more intuitive, it is how to choose the point in the integrand:

Lemma 4.6.2. *Let c be a locally bounded adapted L-process, and X be a continuous semimartingale. Then, with $t_i^n = t \wedge \frac{i}{2^n}$:*

$$I^n(t) := \sum_{i=1}^{\infty} c\left(t_{i-1}^n\right)\left(X_{t_i^n} - X_{t_{i-1}^n}\right) \longrightarrow \int_0^t c(s)\mathrm{d}X_s =: I(t)$$

, the convergence is uniform in probability on compacts, i.e. $\forall\ K > 0$,

$$\sup_{t \le K}\left|I^n(t) - I(t)\right| \xrightarrow{p.} 0$$

Proof. Let $c^n(t) = c(t_{i-1}^n)$ on $(t_{i-1}^n, t_i^n]$, then $\forall\ t, \omega$:

$$c^n(t,\omega) \longrightarrow c(t,\omega)$$

Also,

$$I^n(t) = \int_0^t c^n(s)\mathrm{d}X_s$$

Consider the cannonical decomposition, $X_t = X_0 + M_t + A_t$, for each ω:

$$\int_0^t c^n(s)\mathrm{d}A_s \longrightarrow \int_0^t c(s)\mathrm{d}A_s$$

, which is guaranteed by classical Lebesgue intergation, it remains to show:

$$\int_0^t c^n(s)\mathrm{d}M_s \longrightarrow \int_0^t c(s)\mathrm{d}M_s$$

, which is equivalent: c is bounded, and $M \in \mathcal{M}_0^2$. Then we completed the proof. $\square$

Lemma 4.6.3. *Let X, Y be two continuous semimartingale, define:*

$$S^n(t) = \sum_{i=1}^{\infty} \frac{1}{2}\left(Y_{t_i^n} + Y_{t_{i-1}^n}\right) \cdot \left(X_{t_i^n} - X_{t_{i-1}^n}\right)$$

, and $S(t) = \int_0^t Y_s \partial X_s$, then $\forall$ $K > 0$:

$$\sup_{t \leq K} |S^n(t) - S(t)| \xrightarrow{p.} 0$$

Proof. Notice the following rearrangement:

$$\begin{aligned}
S^n(t) &= \sum_{i=1}^{\infty} \frac{1}{2}\left(Y_{t_i^n} + Y_{t_{i-1}^n}\right) \cdot \left(X_{t_i^n} - X_{t_{i-1}^n}\right) \\
&= \sum_{i=1}^{\infty} \left[Y_{t_{i-1}^n} + \frac{1}{2}\left(Y_{t_i^n} - Y_{t_{i-1}^n}\right)\right] \cdot \left(X_{t_i^n} - X_{t_{i-1}^n}\right) \\
&= \sum_{i=1}^{\infty} Y_{t_{i-1}^n} \cdot \left(X_{t_i^n} - X_{t_{i-1}^n}\right) + \frac{1}{2}\sum_{i=1}^{\infty} \Delta Y_{t_i^n} \Delta X_{t_i^n}
\end{aligned}$$

, where $\Delta Y_{t_i^n} = Y_{t_i^n} - Y_{t_{i-1}^n}$, and $\Delta X_{t_i^n} = X_{t_i^n} - X_{t_{i-1}^n}$. It remains to show:

$$\sum_{i=1}^{\infty} \Delta Y_{t_i^n} \Delta X_{t_i^n} \longrightarrow [X, Y]_t = \frac{1}{4}\left([X+Y]_t - [X-Y]_t\right)$$

Then this is equivalent to show:

$$\sum_{i=1}^{\infty} \left(\Delta X_{t_i^n}\right)^2 = \sum_{i=1}^{\infty} \left(\Delta M_{t_i^n} + \Delta A_{t_i^n}\right)^2 \longrightarrow [X]_t = [M]_t$$

Other left as exercise for reader. □

The two lemmas just given are clearly relevant to the problem of simulating processes described by stochastic differential equations, which we will see later.

Chapter 5

Stochastic Differential Equations